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Ⅰ. 데이터 전처리

Ⅰ-(1) 데이터 클렌징-제공데이터

Product

1. R Code

```
setwd("C:/Users/j3eun/OneDrive/Desktop/lpoint")
product<-read_csv("product.csv")
product[29650,8]<-1440
product[34106,8]<-1387
product<-product%>%mutate(PD_BUY_TOTAL=PD_BUY_AM*PD_BUY_CT)
product<-product%>%filter(HITS_SEQ!=1)
write_rds(product,"product.rds")
read_rds("product.rds")
```

2. 데이터 클렌징 논리적 이유

- 29650열과 34106열의 PD_BUY_CT값이 결측값인 것을 확인. 직접 값을 찾아서 넣어줌.

View(product[29650,])

	CLNT_ID	SESS_ID	HITS_SEQ	PD_C	PD_ADD_NM	PD_BRA_NM	PD_BUY_AM	PD_BUY_CT
1	3007548	03553431	112	643396	원단선택:91번|손잡이선택:파죽|설치공간:천정목재 />	리드센스	100	NA

29651	3007548	3553431	112	643396	원단선택:9	리드센스	100	1,440

View(product[34106,])

	CLNT_ID	SESS_ID	HITS_SEQ	PD_C	PD_ADD_NM	PD_BRA_NM	PD_BUY_AM	PD_BUY_CT
1	5387167	09938956	64	644954	원단선택:68번|손잡이선택:우측|설치공간선택:천정시>	리드센스	100	NA

34107	5387167	9938956	64	644954	원단선택:6	리드센스	100	1,387

- PD_BUY_TOTAL 파생변수 생성

- HITS_SEG가 "1"인 데이터는 분석 시 이상치로 간주하여 제거함.

Search1

1. R Code

```
search1<-read_csv("search1.csv")
search1$CLNT_ID<-as.character(search1$CLNT_ID)
```

```
write_rds(search1,"search1.rds")
read_rds("search1.rds")
```

2. 데이터 클렌징 논리적 이유

- CLNT_ID는 범주형 변수이므로 character 타입으로 바꿈.
- 다른 오류 없음.

Search2

1. R Code

```
search2<-
read_csv("search2.csv",col_types=cols(SEARCH_CNT=col_character(),SESS_DT=col_character()))
search2$SEARCH_CNT<- as.numeric(gsub(",","",search2$SEARCH_CNT))
search2$SESS_DT<-lubridate::as_date(search2$SESS_DT)
search2$SESS_DAY<-weekdays(search2$SESS_DT)
write_rds(search2,"search2.rds")
read_rds("search2.rds")
```

2. 데이터 클렌징 논리적 이유

- 자료 불러올 때 오류 수정과정 포함.
- SESS_DT를 Date형식으로 바꿈.
- SESS_DAY 파생변수 생성 (해당 요일)

Custom

1. R Code

```
custom<-read_csv("custom.csv")
write_rds(custom,"custom.rds")
read_rds("custom.rds")
```

2. 데이터 클렌징 논리적 이유

- Missing Value 없음.

Session

1. R Code

```
session<-read_csv("session.csv",col_types=cols(CLNT_ID=col_character(),SESS_DT=col_character()))
session$TOT_SESS_HR_V<-ifelse(is.na(session$TOT_SESS_HR_V), 0,session$TOT_SESS_HR_V)
session$TOT_PAG_VIEW_CT<-ifelse(is.na(session$TOT_PAG_VIEW_CT), 0,session$TOT_PAG_VIEW_CT)
session$SESS_DT<-lubridate::as_date(session$SESS_DT)
session$SESS_DAY<-weekdays(session$SESS_DT)
write_rds(session,"session.rds")
read_rds("session.rds")
```

2. 데이터 클렌징 논리적 이유

- TOT_SESS_HR_V와 TOT_PAG_VIEW_CT의 결측값을 0으로 대체함.

→ TOT_SESS_HR_V의 NA 값은 세션 내에서 1초 이내 머물렀기 때문에 값이 입력되지 않은 것으로 판단함. TOT_SESS_HR_V가 0인 경우가 없고, TOT_PAG_VIEW_CT 값이 1이면서 TOT_SESS_HR_V 값이 NA인 경우가 약 98% 이기 때문임.

```
> session%>%filter(TOT_SESS_HR_V==0)
# A tibble: 0 x 9
# ... with 9 variables: CLNT_ID <int>, SESS_ID <chr>, SESS_SEQ <int>, SESS_DT <int>, TOT_PAG_VIEW_CT <int>,
#   TOT_SESS_HR_V <dbl>, DVC_CTG_NM <chr>, ZON_NM <chr>, CITY_NM <chr>

> session%>%filter(is.na(TOT_SESS_HR_V))%>%count()
# A tibble: 1 x 1
      n
  <int>
1 14202
> session%>%filter(TOT_PAG_VIEW_CT==1,is.na(TOT_SESS_HR_V))%>%count()
# A tibble: 1 x 1
      n
  <int>
1 13870
>
> 13870/14202
[1] 0.976623
```

→ TOT_PAG_VIEW_CT가 0인 경우도 존재하지 않고, 약 97%가 TOT_SESS_HR_V가 NA일 때 NA값을 보임. TOT_SESS_HR_V가 NA가 아님에도 TOT_PAG_VIEW_CT가 NA인 경우는 세션 오류로 판단함.

```
> session%>%filter(TOT_PAG_VIEW_CT==0)
# A tibble: 0 x 9
# ... with 9 variables: CLNT_ID <int>, SESS_ID <chr>, SESS_SEQ <int>, SESS_DT <int>, TOT_PAG_VIEW_CT <int>,
#   TOT_SESS_HR_V <dbl>, DVC_CTG_NM <chr>, ZON_NM <chr>, CITY_NM <chr>
```

```
> session%>%filter(is.na(TOT_PAG_VIEW_CT),is.na(TOT_SESS_HR_V))%>%count()
# A tibble: 1 x 1
      n
  <int>
1   265
> session%>%filter(is.na(TOT_PAG_VIEW_CT))%>%count()
# A tibble: 1 x 1
      n
  <int>
1   274
> 265/274
[1] 0.9671533
```

- SESS_DT를 Date형식으로 바꿈.

- SESS_DAY 파생변수 생성 (해당 요일)

Master

1. R Code

```
master<-read.csv("master2.csv",header=TRUE,sep=",")
master<-as_tibble(master)
master<-master%>%select(-"X")
master$PD_C<-as.character(master$PD_C)
master[159325,3]<-"가구"
master[159325,4]<-"침실가구"
master[159325,5]<-"장롱"
master[771819,3]<-"패션잡화"
master[771819,4]<-"여성화"
master[771819,5]<-"여성펌프스"
master[817746,3]<-"냉장/세탁가전"
master[817746,4]<-"세탁기"
master[817746,5]<-"건조기"
write_rds(master,"master.rds")
read_rds("master.rds")
```

2. 데이터 클렌징 논리적 이유

- 자료 불러올 때 오류 수정 과정 포함.

- CLAC2_NM 오류 발견.

```
master%>%filter(CLAC2_NM%in%c("가구","냉장/세탁가전","패션잡화"))
```

```
# A tibble: 3 x 5
  PD_C   PD_NM                                                    CLAC1_NM              CLAC2_NM      CLAC3_NM
  <chr>  <fct>                                                     <fct>                 <fct>         <fct>
1 666212 □탬퍼 무료증정□트윈 화이트그레시 슬라이딩 싱글장(1500) ""      가구            침실가구
2 6848   스냅와니 힐 E8310(9cm)                                  " - 블랙 / 225 / 9cm" 패션잡화      여성화
3 186564 [키트미포함] 의류건조기 DV14N8520KP [14KG]            ""                    냉장/세탁가전 세탁기
```

```
> which(master$PD_C=="666212")
[1] 159325
> which(master$PD_C=="6848")
[1] 771819
> which(master$PD_C=="186564")
[1] 817746
```

원래 가구, 냉장/세탁가전, 패션잡화 는 CLAC1_NM에 해당하는 값임. 따라서 CLAC1_NM, CLAC2_NM, CLAC3_NM에 올바른 값을 넣어줌.

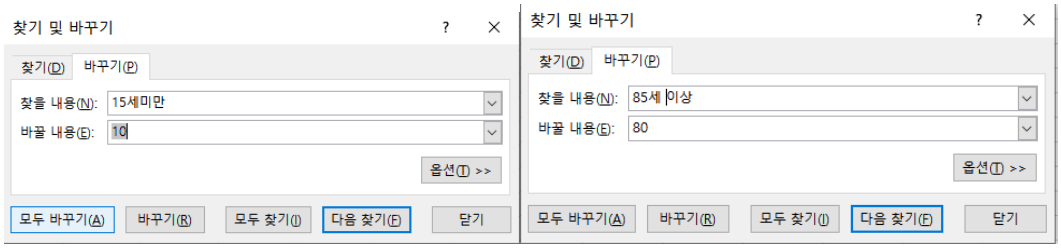
```
> master[c(159325,771819,817746),]
# A tibble: 3 x 5
  PD_C   PD_NM                                                    CLAC1_NM      CLAC2_NM CLAC3_NM
  <chr>  <fct>                                                     <fct>         <fct>    <fct>
1 666212 ★탬퍼 무료증정★트윈 화이트그레시 슬라이딩 싱글장(1500) 가구   침실가구 장롱
2 6848   스냅와니 힐 E8310(9cm)                                   패션잡화      여성화   여성펌프스
3 186564 [키트미포함] 의류건조기 DV14N8520KP [14KG]             냉장/세탁가전 세탁기   건조기
```

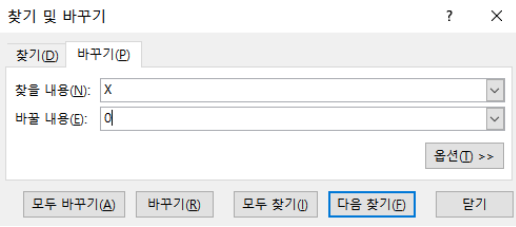
Ⅰ-(1) 데이터 클렌징-외부데이터

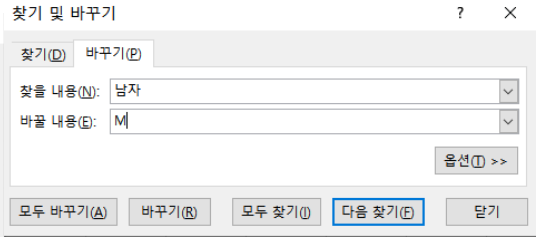
사용한 외부데이터: 1. 1인가구 통계지표 2. 연도별 일·가정 양립제도 인지도
외부데이터 출처: KOSIS 국가 통계포털

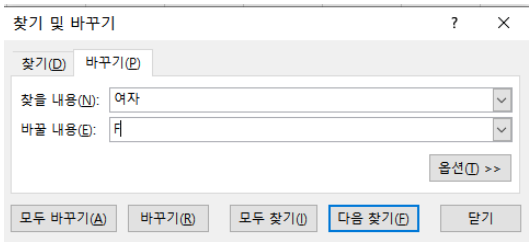
Oneperson_house 정리

Oneperson_house를 엑셀에서 연 다음
15세미만은 10으로,85세이상은 80으로, "찾기및바꾸기"를 사용하여 원자료 가공해줌.









2015년 정리과정

```
op2015<-read.csv("oneperson_2015.csv")
op2015<-as_tibble(op2015)
op2015$REGION<-as.character(op2015$REGION)
op2015$GENDER<-as.character(op2015$GENDER)
op2015$AGE<-as.character(op2015$AGE)
op2015<-
op2015%>%separate(AGE,into=c("AGE","X"),sep=1)%>%select(REGION,GENDER,AGE,ALONE_CNT)
op2015$Y<-0
op2015<-op2015%>%unite(AGE,AGE,Y,sep="")
op2015<-
op2015%>%group_by(REGION,GENDER,AGE)%>%summarize(ALONE_CNT=sum(ALONE_CNT))
op2015<-op2015%>%ungroup()
op2015<-op2015%>%mutate(YEAR=2015)
write_rds(op2015,"op2015.rds")
```

2016년 정리과정

```
op2016<-read.csv("oneperson_2016.csv")
op2016<-as_tibble(op2016)
op2016$REGION<-as.character(op2016$REGION)
op2016$GENDER<-as.character(op2016$GENDER)
op2016$AGE<-as.character(op2016$AGE)
op2016<-
op2016%>%separate(AGE,into=c("AGE","X"),sep=1)%>%select(REGION,GENDER,AGE,ALONE_CNT)
op2016$Y<-0
```

```
op2016<-op2016%>%unite(AGE,AGE,Y,sep="")
op2016<-
op2016%>%group_by(REGION,GENDER,AGE)%>%summarize(ALONE_CNT=sum(ALONE_CNT))
op2016<-op2016%>%ungroup()
op2016<-op2016%>%mutate(YEAR=2016)
write_rds(op2016,"op2016.rds")
```

2017년 정리과정

```
op2017<-read.csv("oneperson_2017.csv")
op2017<-as_tibble(op2016)
op2017$REGION<-as.character(op2017$REGION)
op2017$GENDER<-as.character(op2017$GENDER)
op2017$AGE<-as.character(op2017$AGE)
op2017<-
op2017%>%separate(AGE,into=c("AGE","X"),sep=1)%>%select(REGION,GENDER,AGE,ALONE_CNT)
op2017$Y<-0
op2017<-op2017%>%unite(AGE,AGE,Y,sep="")
op2017<-
op2017%>%group_by(REGION,GENDER,AGE)%>%summarize(ALONE_CNT=sum(ALONE_CNT))
op2017<-op2017%>%ungroup()
op2017<-op2017%>%mutate(YEAR=2017)
write_rds(op2017,"op2017.rds")
```

parenting2013~2015 정리

육아똑똑지수 파생변수 산출방법
가중치: 여자 각각의 "알고있다%"를 총합으로 나눔
가중치를 남자 각각의 "알고있다%"에 곱하여 더함.
<parenting2013.csv정리본>

	A	B	C	D	E	F	G	H
1	TRAITS	출산(전후)휴가제 -	배우자출산휴가제	육아휴직제	육아기 근로시간 단축제	직장보육지원	가족돌봄휴직제	육아똑똑지수
2	M20	75.2	64.6	65.4	27	44.8	22.8	56.32068465
3	F20	81.1	66.4	75.9	40.4	51.2	29.7	
4	M30	82.2	75	74.3	36.6	54.2	26.6	63.89821518
5	F30	86.7	78.3	82.7	50.2	64.5	35.4	
6	M40	80.8	74	75.4	38.6	58	32.2	65.10352304
7	F40	82	71.9	77.4	42.5	59.5	35.7	

Ⅰ-(2) 데이터 탐색–제공데이터

CLAC1_NM별 매출액 순위

product<-read_rds("product.rds")

master<-read_rds("master.rds")

pm<-inner_join(product,master)

pm%>%group_by(CLAC1_NM)%>%summarize(REV_CLAC=sum(PD_BUY_TOTAL))%>%arrange(desc(

REV_CLAC))

```
# A tibble: 37 x 2
   CLAC1_NM           REV_CLAC
   <fct>                 <dbl>
 1 패션잡화         41328767091
 2 여성의류         29912477428
 3 스포츠패션       26752614430
 4 화장품/뷰티케어 26234397616
 5 남성의류         16519760725
 6 가구              11110457097
 7 생활/주방가전     9555401273
 8 건강식품          8949252556
 9 유아동의류        8081599950
10 냉장/세탁가전     6998907122
# ... with 27 more rows
```

지역별 매출액과 구매횟수 순위

session<-read_rds("session.rds")

ps<-inner_join(product, session)

ps_region<-ps%>%group_by(ZON_NM)%>%summarize(REV_REGION=sum(PD_BUY_TOTAL))

ps_region%>%arrange(desc(REV_REGION))

```
# A tibble: 16 x 2
   ZON_NM              REV_REGION
   <chr>                    <dbl>
 1 Seoul              106075046975
 2 Gyeonggi-do         46428793046
 3 Busan               30585815914
 4 Gyeongsangnam-do    11853095649
 5 Daegu               11051716444
 6 Incheon              9562448800
 7 Gyeongsangbuk-do     8579229842
 8 Jeollabuk-do         7687206005
 9 Chungcheongnam-do    7318032504
10 Gwangju              6784636987
11 Daejeon              5814397091
12 Jeollanam-do         4752923740
13 Ulsan                4419702450
14 Chungcheongbuk-do    4293291514
15 Gangwon-do           4087342939
16 Jeju-do              2144365880
```

ps_region_trans<-ps%>%group_by(ZON_NM)%>%summarize(n=n())

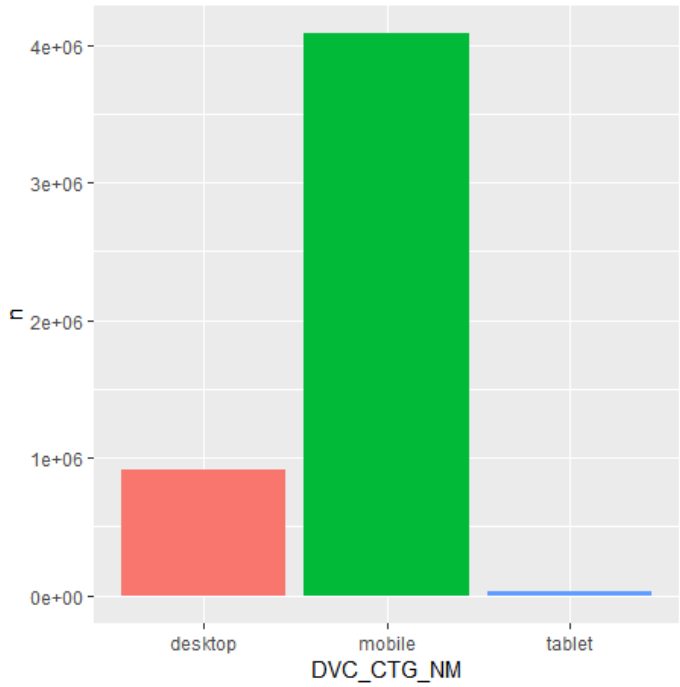
ps_region_trans%>%arrange(desc(n))

```
# A tibble: 16 x 2
   ZON_NM                  n
   <chr>               <int>
 1 Seoul             1820948
 2 Gyeonggi-do        873269
 3 Busan              549342
 4 Gyeongsangnam-do   247647
 5 Daegu              215533
 6 Incheon            185660
 7 Gyeongsangbuk-do   175716
 8 Jeollabuk-do       155900
 9 Chungcheongnam-do  151521
10 Gwangju            136024
11 Daejeon            114103
12 Jeollanam-do        97780
13 Ulsan               87852
14 Chungcheongbuk-do   87359
15 Gangwon-do          81925
16 Jeju-do             39151
```

기기별 매출액 구매횟수 순위

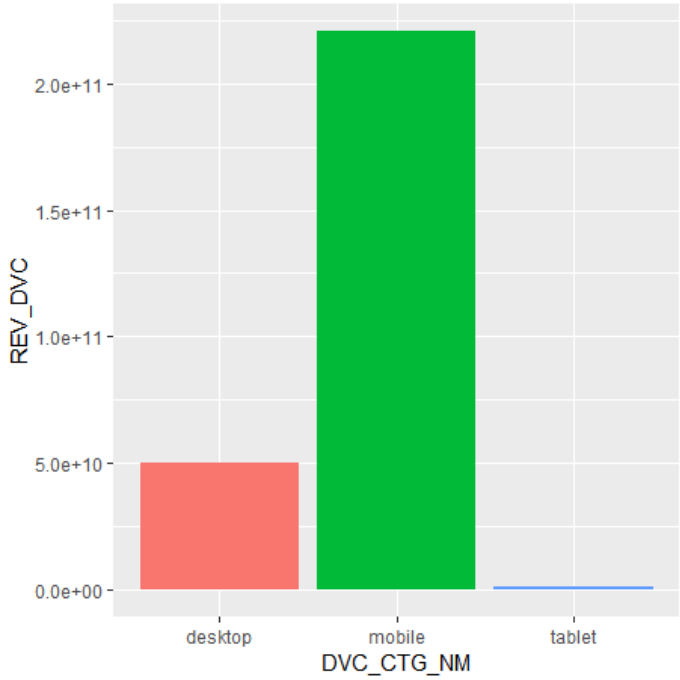
ps_dvc<-ps%>%group_by(DVC_CTG_NM)%>%summarize(n=n())

ps_dvc%>%ggplot(aes(DVC_CTG_NM,n,fill=DVC_CTG_NM))+geom_bar(stat="identity")+guides(fill=FALSE)



ps_dvc_rev<-ps%>%group_by(DVC_CTG_NM)%>%summarise(REV_DVC=sum(PD_BUY_TOTAL))

ps_dvc_rev%>%ggplot(aes(DVC_CTG_NM,REV_DVC,fill=DVC_CTG_NM))+geom_bar(stat="identity")+guides(fill=FALSE)



모바일을 쓰는 고객 중에서 매출이 가장 높은 성별 나이대?

```
custom<-read_rds("custom.rds")
psc<-inner_join(ps,custom)
psc%>%filter(DVC_CTG_NM=="mobile")%>%group_by(CLNT_GENDER,CLNT_AGE)%>%summarize(REV_DVC_GA=sum(PD_BUY_TOTAL))%>%arrange(desc(REV_DVC_GA))
```

```
# A tibble: 16 x 3
# Groups:   CLNT_GENDER [2]
   CLNT_GENDER CLNT_AGE  REV_DVC_GA
   <chr>          <dbl>       <dbl>
 1 F                 30 77062956670
 2 F                 40 68928194920
 3 F                 20 17772157340
 4 M                 40 12368268170
 5 F                 50 12059853150
 6 M                 30 11652446470
 7 M                 50  3124370730
 8 M                 20  2953046980
 9 F                 60  2406945780
10 M                 60   495460900
11 F                 10   441211980
12 F                 70   188070450
13 M                 10   142357300
14 M                 70    78105630
15 F                 80     9552750
16 M                 80     3291400
```

성별 나이대별 매출액

```
pc<-inner_join(product,custom)
pc%>%group_by(CLNT_GENDER,CLNT_AGE)%>%summarize(REV_GA=sum(PD_BUY_TOTAL))%>%arr
ange(desc(REV_GA))
# A tibble: 16 x 3
# Groups:   CLNT_GENDER [2]
   CLNT_GENDER CLNT_AGE      REV_GA
   <chr>          <dbl>       <dbl>
 1 F                 30 77274160100
 2 F                 40 69289438960
 3 F                 20 17836310510
 4 M                 40 12609639740
 5 F                 50 12177132020
 6 M                 30 11703598871
 7 M                 50  3157926380
 8 M                 20  2966230490
 9 F                 60  2434191720
10 M                 60   503644390
11 F                 10   444122800
12 F                 70   190678580
13 M                 10   145106640
14 M                 70    81293510
15 F                 80     9955270
16 M                 80     3291400
```

*** 외부데이터의 자세한 탐색과정은 "서비스제안"파일에서 다룰 예정.**

Ⅱ. 온라인 선호지수 개발

사용한 패키지: arules package / 사용한 Function: apriori()

개발 개요: 연관규칙분석(Association Rule)를 이용한 온라인 선호지수 개발 (Apriori algorithm)

Ⅱ-(1) 주요 상품 군 설정 단계

- 대용량 데이터로부터 효율적으로 연관규칙(association rules)을 도출할 수 있는 apriori algorithm 사용.
- Master 데이터에 있는 CLAC1_NM(상품 대분류명) 간의 전체 연관규칙 순위(IS측도)를 기준으로 주요 상품 군 설정.

Support(지지도) pruning 과정

1. R Code

```
library(arules)

pm1<-
left_join(product,master)%>%select(CLNT_ID,SESS_ID,CLAC1_NM)%>%unique()%>%filter(!is.na(CLAC1_NM))

pm1<-pm1%>%group_by(CLNT_ID,SESS_ID)%>% unite(CLNT,CLNT_ID,SESS_ID,sep="/")

write.csv(pm1,"pm1.csv")

pm1_trans<-
read.transactions(file="pm1.csv",format="single",sep=",",cols=c("CLNT","CLAC1_NM"),rm.duplicates=T)

pm_rule1<-apriori(pm1_trans,parameter=list(support=0.001,confidence=0.02,minlen=2))

inspect(sort(pm_rule1, by = "support")[1:20])
```

```
     lhs                    rhs                 support     confidence lift      count
[1]  {남성의류}          => {여성의류}          0.010568783 0.10238975 0.7021615 26637
[2]  {여성의류}          => {남성의류}          0.010568783 0.07247789 0.7021615 26637
[3]  {남성의류}          => {스포츠패션}        0.007562054 0.07326074 0.5195299 19059
[4]  {스포츠패션}        => {남성의류}          0.007562054 0.05362645 0.5195299 19059
[5]  {패션잡화}          => {여성의류}          0.007358511 0.05770559 0.3957295 18546
[6]  {여성의류}          => {패션잡화}          0.007358511 0.05046270 0.3957295 18546
[7]  {패션잡화}          => {스포츠패션}        0.006852628 0.05373845 0.3810872 17271
[8]  {스포츠패션}        => {패션잡화}          0.006852628 0.04859554 0.3810872 17271
[9]  {퍼스널케어}        => {화장품/뷰티케어}   0.005381402 0.15132041 1.0836012 13563
[10] {화장품/뷰티케어}   => {퍼스널케어}        0.005381402 0.03853607 1.0836012 13563
[11] {스포츠패션}        => {여성의류}          0.005375847 0.03812292 0.2614368 13549
[12] {여성의류}          => {스포츠패션}        0.005375847 0.03686612 0.2614368 13549
[13] {남성의류}          => {패션잡화}          0.004662850 0.04517342 0.3542509 11752
[14] {패션잡화}          => {남성의류}          0.004662850 0.03656617 0.3542509 11752
[15] {유아동의류}        => {스포츠패션}        0.004596193 0.08469879 0.6006431 11584
[16] {스포츠패션}        => {유아동의류}        0.004596193 0.03259398 0.6006431 11584
[17] {유아동의류}        => {속옷/양말/홈웨어}  0.004398601 0.08105757 1.1348928 11086
[18] {속옷/양말/홈웨어} => {유아동의류}        0.004398601 0.06158513 1.1348928 11086
[19] {속옷/양말/홈웨어} => {여성의류}          0.004378365 0.06130181 0.4203914 11035
[20] {여성의류}          => {속옷/양말/홈웨어} 0.004378365 0.03002566 0.4203914 11035
```

2. 코드의 논리적 과정 설명

- product 자료와 master자료를 병합하여 중복되는 열은 unique()함수를 써서 제거함. 또한 CLAC1_NM이 NA인 값은 제거함.

- 고객이 한 세션 내에서 구매하는 상품을 보는 것이므로 CLNT_ID 와 SESS_ID를 그룹화해서 분석. (분석에 용이하게 두 변수를 unite하여 CLNT라는 변수로 만듦.)

- read.transaction()함수를 사용하여 연관규칙분석에 사용할 데이터 셋 형태를 맞춰줌. (rm.duplicate=T를 써서 중복된 것은 제거함. 동일한 고객, 세션, 상품은 하나의 거래로 생각했기 때문임)

- CLAC1_NM의 연관규칙을 최대한 다양하게 보기 위해 support를 0.001로 낮추어 나타내어 봄. (너무 낮은 support는 비중이 적은 것이므로 0.001이하는 pruning함.) 그 후 support의 내림차순으로 print함.

IS Measure 순으로 주요 상품 군 결정

1. R Code

```
pm_rule1_df<-as(pm_rule1,"data.frame")

pm_rule1_df<-transform(pm_rule1_df,IS=sqrt(lift*support))

pm_rule1_df[order(-pm_rule1_df$IS),][1:20,]
```

```
                                  rules     support confidence      lift count         IS
5        {주방잡화} => {식기/조리기구} 0.001476781 0.11796400 6.5173879  3722 0.09810583
6        {식기/조리기구} => {주방잡화} 0.001476781 0.08159060 6.5173879  3722 0.09810583
51             {남성의류} => {여성의류} 0.010568783 0.10238975 0.7021615 26637 0.08614518
52             {여성의류} => {남성의류} 0.010568783 0.07247789 0.7021615 26637 0.08614518
23   {퍼스널케어} => {화장품/뷰티케어} 0.005381402 0.15132041 1.0836012 13563 0.07636291
24   {화장품/뷰티케어} => {퍼스널케어} 0.005381402 0.03853607 1.0836012 13563 0.07636291
26 {유아동의류} => {속옷/양말/홈웨어} 0.004398601 0.08105757 1.1348928 11086 0.07065366
27 {속옷/양말/홈웨어} => {유아동의류} 0.004398601 0.06158513 1.1348928 11086 0.07065366
49           {남성의류} => {스포츠패션} 0.007562054 0.07326074 0.5195299 19059 0.06267945
50           {스포츠패션} => {남성의류} 0.007562054 0.05362645 0.5195299 19059 0.06267945
3       {세제/위생} => {출산/육아용품} 0.001212135 0.09221806 2.5317697  3055 0.05539716
4       {출산/육아용품} => {세제/위생} 0.001212135 0.03327814 2.5317697  3055 0.05539716
57             {패션잡화} => {여성의류} 0.007358511 0.05770559 0.3957295 18546 0.05396276
58             {여성의류} => {패션잡화} 0.007358511 0.05046270 0.3957295 18546 0.05396276
33         {유아동의류} => {스포츠패션} 0.004596193 0.08469879 0.6006431 11584 0.05254209
34         {스포츠패션} => {유아동의류} 0.004596193 0.03259398 0.6006431 11584 0.05254209
55           {패션잡화} => {스포츠패션} 0.006852628 0.05373845 0.3810872 17271 0.05110234
56           {스포츠패션} => {패션잡화} 0.006852628 0.04859554 0.3810872 17271 0.05110234
30           {유아동의류} => {패션잡화} 0.003882402 0.07154504 0.5610576  9785 0.04667174
31           {패션잡화} => {유아동의류} 0.003882402 0.03044588 0.5610576  9785 0.04667174
```

여기 있는 CLAC1 상품 군들을 주요 상품들이라 판단하고 분류함. 또한 이중에는 없지만 매출액이 높은 상품군도 주의 깊게 봐야 한다고 판단함. 따라서 선호지수를 산출할 때 이를 포함하여 주요 상품 군들을 만듦. (TOP 16)

=> 건강식품, 구기/필드스포츠, 남성의류, 세제/위생, 속옷/양말/홈웨어, 스포츠패션, 시즌스포츠, 식기/조리기구, 아웃도어/레저, 여성의류, 유아동의류, 주방잡화, 출산/육아용품, 패션잡화, 퍼스널케어, 화장품/뷰티케어

- TOP16을 비슷한 부류 끼리 묶어 6개의 주요 상품 군을 설정함.

주요 상품 군	상품 군 구성(16개)
1의류	여성의류 스포츠패션 남성의류 유아동의류 속옷/양말/홈웨어
2스포츠	시즌스포츠 구기/필드스포츠 아웃도어/레저
3기타패션	패션잡화
4뷰티/건강	퍼스널케어, 화장품/뷰티케어 건강식품
5출산/육아	출산/육아용품
6홈케어	식기/조리기구 주방잡화 세제/위생

6개의 주요 상품 군별 CLAC3_NM 연관규칙분석 준비과정

- 각 주요 상품 군의 가장 작은 분류인 상품 소분류명(CLAC3_NM)간의 연관규칙을 분석할 예정.

1. R Code

pm<-left_join(product,master)%>%filter(CLAC1_NM%in%c("건강식품","구기/필드스포츠","남성의류","세제/위생","속옷/양말/홈웨어","스포츠패션","시즌스포츠","식기/조리기구","아웃도어/레저","여성의류","유아동의류","주방잡화","출산/육아용품","패션잡화","퍼스널케어","화장품/뷰티케어"))

pm3<-pm%>%select(CLNT_ID,SESS_ID,CLAC1_NM,CLAC3_NM)

pm3<-pm3%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("여성의류" = "1의류-여성의류", "스포츠패션" = "1의류-스포츠패션", "남성의류" = "1의류-남성의류","유아동의류"="1의류-유아동의류","속옷/양말/홈웨어"="1의류-속옷/양말/홈웨어")))

pm3<-pm3%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("시즌스포츠" = "2스포츠-시즌스포츠", "구기/필드스포츠" = "2스포츠-구기/필드스포츠", "아웃도어/레저" = "2스포츠-아웃도어/레저")))

pm3<-pm3%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("건강식품" = "4뷰티건강-건강식품", "퍼스널케어" = "4뷰티건강-퍼스널케어", "화장품/뷰티케어" = "4뷰티건강-화장품/뷰티케어","출산/육아용품" = "5출산육아-출산/육아용품")))

pm3<-pm3%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("패션잡화" = "3기타패션-패션잡화", "식기/조리기구" = "6홈케어-식기/조리기구", "주방잡화" = "6홈케어-주방잡화","세제/위생"="6홈케어-세제/위생")))

```
# A tibble: 3,987,262 x 4
   CLNT_ID SESS_ID  CLAC1_NM                  CLAC3_NM
   <chr>   <chr>    <chr>                     <fct>
 1 4139680 07605037 4뷰티건강-화장품/뷰티케어 블러셔/쉐이딩/하이라이터
 2 4140076 10189797 1의류-스포츠패션          남성스포츠티셔츠
 3 4142395 06158159 4뷰티건강-퍼스널케어      헤어케어선물세트
 4 4144914 07935714 1의류-유아동의류          여아레깅스
 5 4144917 06406509 4뷰티건강-화장품/뷰티케어 스킨케어세트
 6 4147643 04050743 3기타패션-패션잡화        여성선글라스
 7 4147739 06881895 2스포츠-아웃도어/레저     등산화
 8 4148106 06537318 1의류-스포츠패션          남성스포츠티셔츠
 9 4148129 03942102 4뷰티건강-화장품/뷰티케어 메이크업베이스/프라이머
10 4148527 05718021 1의류-스포츠패션          스포츠선글라스
# ... with 3,987,252 more rows
```

CLAC1_NM의 각 대분류명 앞에 "상품분류"를 따로 이름 붙임. -> 나중에 주요 상품 군 별로

subset할때(%pin%), 분류 별로 이름 하나만 치면 나오게끔 만듦.

```
pm3<-
pm3%>%select(CLNT_ID,SESS_ID,CLAC1_NM,CLAC3_NM)%>%unite(CLAC3_RNM,CLAC1_NM,CLAC3_NM,sep="-")%>%unique()%>%filter(!is.na(CLAC3_RNM))%>%group_by(CLNT_ID,SESS_ID)%>%unite(CLNT,CLNT_ID,SESS_ID,sep="/")

write.csv(pm3,"pm3.csv")

pm3_trans<-
read.transactions(file="pm3.csv",format="single",sep=",",cols=c("CLNT","CLAC3_RNM"),rm.duplicates=T)

pm3_rule<-apriori(pm3_trans,parameter=list(support=0.0001,confidence=0.01,minlen=2))
```

2. 코드의 논리적 과정 설명

- 중복되는 열은 unique()함수를 써서 제거함. 또한 CLAC3_NM이 NA인 값은 제거함.

- CLAC3_NM이 어디 상품 군인지 알기 쉽게 CLAC1_NM과 unite하여 CLAC3_RNM이라는 변수명으로 바꿈..

- 고객이 한 세션 내에서 구매하는 상품을 보는 것이므로 CLNT_ID 와 SESS_ID를 그룹화해서 분석. (분석에 용이하게 두 변수를 unite하여 CLNT라는 변수로 만듦.)

- read.transaction()함수를 사용하여 연관규칙분석에 사용할 데이터 셋 형태를 맞춰줌. (rm.duplicate=T를 써서 중복된 것은 제거함. 동일한 고객, 세션, 상품은 하나의 거래로 생각했기 때문임)

- CLAC3_RNM의 연관규칙을 최대한 모든 주요 상품 군들이 나오게끔 support를 0.0001로 낮추어 나타내어 봄. 그 후 각각의 주요상품군만 subset하여 연관규칙을 분석할 예정.

*참고사항: 연관규칙분석 결과를 일일이 보며 Explainable한 값들만 찾아 ppt 작성함. 그러나 코드 리스트에는 모든 관측 값을 첨부하기에 양이 많으므로 적당량만 붙임.

1의류(Clothing)의 연관규칙분석

rule_clothing<-subset(pm3_rule,items %pin% c("1"))

inspect(sort(rule_clothing,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                  rhs                                  support     confidence lift      count
[1]  {1의류-남성의류-남성캐주얼바지}     => {1의류-남성의류-남성티셔츠}        0.007969856 0.25663442  3.719146 17378
[2]  {1의류-남성의류-남성티셔츠}         => {1의류-남성의류-남성캐주얼바지}    0.007969856 0.11549914  3.719146 17378
[3]  {1의류-여성의류-여성남방셔츠}       => {1의류-여성의류-여성티셔츠/탑}     0.004793012 0.13032310  3.475983 10451
[4]  {1의류-여성의류-여성티셔츠/탑}      => {1의류-여성의류-여성남방셔츠}      0.004793012 0.12783941  3.475983 10451
[5]  {1의류-여성의류-여성바지}           => {1의류-여성의류-여성티셔츠/탑}      0.004184885 0.15361436  4.097208  9125
[6]  {1의류-여성의류-여성티셔츠/탑}      => {1의류-여성의류-여성바지}           0.004184885 0.11161943  4.097208  9125
[7]  {1의류-여성의류-여성남방셔츠}       => {1의류-여성의류-여성원피스}         0.004022993 0.10938611  2.265465  8772
[8]  {1의류-여성의류-여성원피스}         => {1의류-여성의류-여성남방셔츠}       0.004022993 0.08331909  2.265465  8772
[9]  {1의류-여성의류-여성바지}           => {1의류-여성의류-여성남방셔츠}       0.004022535 0.14765496  4.014772  8771
[10] {1의류-여성의류-여성남방셔츠}       => {1의류-여성의류-여성바지}           0.004022535 0.10937364  4.014772  8771
[11] {1의류-여성의류-여성티셔츠/탑}      => {1의류-여성의류-여성원피스}         0.003584096 0.09559516  1.979845  7815
[12] {1의류-여성의류-여성원피스}         => {1의류-여성의류-여성티셔츠/탑}      0.003584096 0.07422921  1.979845  7815
[13] {1의류-여성의류-여성티셔츠/탑}      => {1의류-남성의류-남성티셔츠}         0.003138320 0.08370540  1.213058  6843
[14] {1의류-남성의류-남성티셔츠}         => {1의류-여성의류-여성티셔츠/탑}      0.003138320 0.04548053  1.213058  6843
[15] {1의류-유아동의류-남아바지}         => {1의류-유아동의류-영유아티셔츠/탑} 0.002819122 0.25133908 18.583172  6147
[16] {1의류-유아동의류-영유아티셔츠/탑} => {1의류-유아동의류-남아바지}         0.002819122 0.20843647 18.583172  6147
[17] {1의류-여성의류-여성스커트}         => {1의류-여성의류-여성남방셔츠}       0.002659065 0.17273946  4.696825  5798
[18] {1의류-여성의류-여성남방셔츠}       => {1의류-여성의류-여성스커트}         0.002659065 0.07230057  4.696825  5798
[19] {1의류-여성의류-여성바지}           => {1의류-여성의류-여성원피스}          0.002491669 0.09146157  1.894235  5433
[20] {1의류-여성의류-여성원피스}         => {1의류-여성의류-여성바지}            0.002491669 0.05160426  1.894235  5433
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_clothing_df<-as(rule_clothing,"data.frame")

rule_clothing_df<-transform(rule_clothing_df,IS=sqrt(lift*support))

rule_clothing_df[order(-rule_clothing_df$IS),]

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2스포츠(Sports)의 연관규칙분석

rule_sports<-subset(pm3_rule, items %pin% c("2"))

inspect(sort(rule_sports,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                          rhs                                          support      confidence lift       count
[1]  {2스포츠-시즌스포츠-남성비치웨어}          => {2스포츠-시즌스포츠-여성비치웨어}         0.0008594493 0.17791702 26.7435557 1874
[2]  {2스포츠-시즌스포츠-여성비치웨어}          => {2스포츠-시즌스포츠-남성비치웨어}         0.0008594493 0.12918792 26.7435557 1874
[3]  {2스포츠-시즌스포츠-여성비치웨어}          => {2스포츠-시즌스포츠-남성수영복}           0.0007126917 0.10712808 14.5447788 1554
[4]  {2스포츠-시즌스포츠-남성수영복}            => {2스포츠-시즌스포츠-여성비치웨어}         0.0007126917 0.09676214 14.5447788 1554
[5]  {2스포츠-시즌스포츠-남성비치웨어}          => {2스포츠-시즌스포츠-남성수영복}           0.0007085641 0.14668186 19.9149939 1545
[6]  {2스포츠-시즌스포츠-남성수영복}            => {2스포츠-시즌스포츠-남성비치웨어}         0.0007085641 0.09620174 19.9149939 1545
[7]  {2스포츠-시즌스포츠-남성수영복}            => {1의류-남성의류-남성티셔츠}                0.0004297247 0.05834371  0.8455169  937
[8]  {2스포츠-시즌스포츠-남성수영복}            => {2스포츠-시즌스포츠-아동수영복}           0.0003948697 0.05361146  5.2954908  861
[9]  {2스포츠-시즌스포츠-아동수영복}            => {2스포츠-시즌스포츠-남성수영복}           0.0003948697 0.03900340  5.2954908  861
[10] {2스포츠-구기/필드스포츠-골프패션잡화}    => {2스포츠-구기/필드스포츠-골프필드용품}    0.0003866146 0.12555853 27.7973515  843
[11] {2스포츠-구기/필드스포츠-골프필드용품}    => {2스포츠-구기/필드스포츠-골프패션잡화}    0.0003866146 0.08559245 27.7973515  843
[12] {2스포츠-시즌스포츠-여성비치웨어}          => {2스포츠-시즌스포츠-아동수영복}           0.0003370839 0.05066869  5.0048178  735
[13] {2스포츠-시즌스포츠-아동수영복}            => {2스포츠-시즌스포츠-여성비치웨어}         0.0003370839 0.03329558  5.0048178  735
[14] {2스포츠-시즌스포츠-수영가방}              => {2스포츠-시즌스포츠-아동수영복}           0.0003040634 0.15910727 15.7158775  663
[15] {2스포츠-시즌스포츠-아동수영복}            => {2스포츠-시즌스포츠-수영가방}             0.0003040634 0.03003398 15.7158775  663
[16] {2스포츠-시즌스포츠-아루아슈즈}            => {1의류-스포츠패션-남성스포츠샌들/슬리퍼} 0.0003040634 0.05151115  4.0649383  663
[17] {1의류-스포츠패션-남성스포츠샌들/슬리퍼} => {2스포츠-시즌스포츠-아루아슈즈}           0.0003040634 0.02399479  4.0649383  663
[18] {2스포츠-시즌스포츠-수영모자}              => {2스포츠-시즌스포츠-여성수영복}           0.0002912222 0.23049002 46.4358910  635
[19] {2스포츠-시즌스포츠-여성수영복}            => {2스포츠-시즌스포츠-수영모자}             0.0002912222 0.05867135 46.4358910  635
[20] {2스포츠-시즌스포츠-아루아슈즈}            => {3기타패션-패션잡화-유아동샌들}           0.0002788395 0.04723798  5.8261668  608
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_sports_df<-as(rule_sports,"data.frame")

rule_sports_df<-transform(rule_sports_df,IS=sqrt(lift*support))

rule_sports_df[order(-rule_sports_df$IS),]

```
                                                                                                                  rules      support confidence        lift count         IS
118                                                  {2스포츠-시즌스포츠-수영모자} => {2스포츠-시즌스포츠-물안경} 0.0002779222 0.21996370 140.8171974   606 0.197828793
119                                                  {2스포츠-시즌스포츠-물안경} => {2스포츠-시즌스포츠-수영모자} 0.0002779222 0.17792132 140.8171974   606 0.197828793
687                                        {2스포츠-시즌스포츠-남성비치웨어} => {2스포츠-시즌스포츠-여성비치웨어} 0.0008594493 0.17791702  26.7435557  1874 0.151607158
688                                        {2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-남성비치웨어} 0.0008594493 0.12918792  26.7435557  1874 0.151607158
689                                          {2스포츠-시즌스포츠-남성비치웨어} => {2스포츠-시즌스포츠-남성수영복} 0.0007085641 0.14668186  19.9149939  1545 0.118789942
690                                          {2스포츠-시즌스포츠-남성수영복} => {2스포츠-시즌스포츠-남성비치웨어} 0.0007085641 0.09620174  19.9149939  1545 0.118789942
122                                            {2스포츠-시즌스포츠-수영모자} => {2스포츠-시즌스포츠-여성수영복} 0.0002912222 0.23049002  46.4358910   635 0.116289123
123                                            {2스포츠-시즌스포츠-여성수영복} => {2스포츠-시즌스포츠-수영모자} 0.0002912222 0.05867135  46.4358910   635 0.116289123
5                                          {2스포츠-아웃도어/레저-캠핑취사} => {2스포츠-아웃도어/레저-기타캠핑용품} 0.0001008959 0.17585931 131.1406470   220 0.115028470
6                                          {2스포츠-아웃도어/레저-기타캠핑용품} => {2스포츠-아웃도어/레저-캠핑취사} 0.0001008959 0.07523940 131.1406470   220 0.115028470
412                                {2스포츠-구기/필드스포츠-골프패션잡화} => {2스포츠-구기/필드스포츠-골프필드용품} 0.0003866146 0.12555853  27.7973515   843 0.103667073
413                                {2스포츠-구기/필드스포츠-골프필드용품} => {2스포츠-구기/필드스포츠-골프패션잡화} 0.0003866146 0.08559245  27.7973515   843 0.103667073
868                                          {2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-남성수영복} 0.0007126917 0.10712808  14.5447788  1554 0.101813275
869                                          {2스포츠-시즌스포츠-남성수영복} => {2스포츠-시즌스포츠-여성비치웨어} 0.0007126917 0.09676214  14.5447788  1554 0.101813275
158                                            {2스포츠-시즌스포츠-물안경} => {2스포츠-시즌스포츠-기타물놀이용품} 0.0002320605 0.14856136  40.9316401   506 0.097460845
159                                            {2스포츠-시즌스포츠-기타물놀이용품} => {2스포츠-시즌스포츠-물안경} 0.0002320605 0.06393733  40.9316401   506 0.097460845
160                                                {2스포츠-시즌스포츠-물안경} => {2스포츠-시즌스포츠-여성수영복} 0.0002215123 0.14180857  28.5695992   483 0.079551977
161                                                {2스포츠-시즌스포츠-여성수영복} => {2스포츠-시즌스포츠-물안경} 0.0002215123 0.04462718  28.5695992   483 0.079551977
120                                            {2스포츠-시즌스포츠-수영모자} => {2스포츠-시즌스포츠-기타물놀이용품} 0.0001696885 0.13430127  37.0026982   370 0.079239714
121                                            {2스포츠-시즌스포츠-기타물놀이용품} => {2스포츠-시즌스포츠-수영모자} 0.0001696885 0.04675259  37.0026982   370 0.079239714
1393 {2스포츠-시즌스포츠-남성수영복,2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-남성비치웨어} 0.0001302474 0.18275418  37.8324582   284 0.070196715
221                                            {2스포츠-시즌스포츠-수영가방} => {2스포츠-시즌스포츠-아동수영복} 0.0003040634 0.15910727  15.7158775   663 0.069127591
222                                            {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-수영가방} 0.0003040634 0.03003398  15.7158775   663 0.069127591
485                                        {2스포츠-시즌스포츠-기타물놀이용품} => {2스포츠-시즌스포츠-여성수영복} 0.0002728774 0.07518322  15.1468589   595 0.064290250
486                                        {2스포츠-시즌스포츠-여성수영복} => {2스포츠-시즌스포츠-기타물놀이용품} 0.0002728774 0.05497552  15.1468589   595 0.064290250
1392 {2스포츠-시즌스포츠-남성비치웨어,2스포츠-시즌스포츠-남성수영복} => {2스포츠-시즌스포츠-여성비치웨어} 0.0001302474 0.18381877  27.6306755   284 0.059990193
49                                 {2스포츠-구기/필드스포츠-골프연습장비} => {2스포츠-구기/필드스포츠-골프필드용품} 0.0001320819 0.11561622  25.5962263   288 0.058144623
50                                 {2스포츠-구기/필드스포츠-골프필드용품} => {2스포츠-구기/필드스포츠-골프연습장비} 0.0001320819 0.02924155  25.5962263   288 0.058144623
73                                         {2스포츠-시즌스포츠-아동비치웨어} => {2스포츠-시즌스포츠-아동수영복} 0.0001871160 0.17824377  17.6060924   408 0.057396698
74                                         {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-아동비치웨어} 0.0001871160 0.01848245  17.6060924   408 0.057396698
544                                  {1의류-스포츠패션-여성골프티셔츠/탑} => {2스포츠-구기/필드스포츠-골프필드용품} 0.0002325191 0.06087165  13.4763499   507 0.055977753
545                                  {2스포츠-구기/필드스포츠-골프필드용품} => {1의류-스포츠패션-여성골프티셔츠/탑} 0.0002325191 0.05147731  13.4763499   507 0.055977753
410                                  {2스포츠-구기/필드스포츠-골프패션잡화} => {1의류-스포츠패션-여성골프티셔츠/탑} 0.0001820712 0.05913018  15.4798100   397 0.053088861
411                                  {1의류-스포츠패션-여성골프티셔츠/탑} => {2스포츠-구기/필드스포츠-골프패션잡화} 0.0001820712 0.04766479  15.4798100   397 0.053088861
1391 {2스포츠-시즌스포츠-남성비치웨어,2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-남성수영복} 0.0001302474 0.15154749  20.5756011   284 0.051767927
165                                            {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-물안경} 0.0001967469 0.01943375  12.4411709   429 0.049474865
164                                            {2스포츠-시즌스포츠-물안경} => {2스포츠-시즌스포츠-아동수영복} 0.0001967469 0.12595420  12.4411709   429 0.049474865
917                                        {2스포츠-시즌스포츠-남성수영복} => {2스포츠-시즌스포츠-아동수영복} 0.0003948697 0.05361146   5.2954908   861 0.045727770
918                                        {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-남성수영복} 0.0003948697 0.03900340   5.2954908   861 0.045727770
126                                            {2스포츠-시즌스포츠-수영모자} => {2스포츠-시즌스포츠-아동수영복} 0.0001632679 0.12921960  12.7637122   356 0.045649796
127                                            {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-수영모자} 0.0001632679 0.01612684  12.7637122   356 0.045649796
491                                        {2스포츠-시즌스포츠-기타물놀이용품} => {2스포츠-시즌스포츠-남성수영복} 0.0002306846 0.06355825   8.6293030   503 0.044616674
492                                        {2스포츠-시즌스포츠-남성수영복} => {2스포츠-시즌스포츠-기타물놀이용품} 0.0002306846 0.03132005   8.6293030   503 0.044616674
219                                            {2스포츠-시즌스포츠-수영가방} => {2스포츠-시즌스포츠-여성비치웨어} 0.0001582231 0.08279338  12.4450670   345 0.044374504
220                                            {2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-수영가방} 0.0001582231 0.02378326  12.4450670   345 0.044374504
213                                    {1의류-스포츠패션-여성골프바지} => {2스포츠-구기/필드스포츠-골프필드용품} 0.0001316232 0.06620531  14.6571650   287 0.043922927
214                                    {2스포츠-구기/필드스포츠-골프필드용품} => {1의류-스포츠패션-여성골프바지} 0.0001316232 0.02914001  14.6571650   287 0.043922927
873                                          {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-여성비치웨어} 0.0003370839 0.03329558   5.0048178   735 0.041073636
872                                          {2스포츠-시즌스포츠-여성비치웨어} => {2스포츠-시즌스포츠-아동수영복} 0.0003370839 0.05066869   5.0048178   735 0.041073636
737                                            {2스포츠-시즌스포츠-아루아슈즈} => {3기타패션-패션잡화-유아동샌들} 0.0002788395 0.04723798   5.8261668   608 0.040305897
738                                            {3기타패션-패션잡화-유아동샌들} => {2스포츠-시즌스포츠-아루아슈즈} 0.0002788395 0.03439109   5.8261668   608 0.040305897
493                                        {2스포츠-시즌스포츠-기타물놀이용품} => {2스포츠-시즌스포츠-아동수영복} 0.0002439845 0.06722264   6.6399406   532 0.040249756
494                                        {2스포츠-시즌스포츠-아동수영복} => {2스포츠-시즌스포츠-기타물놀이용품} 0.0002439845 0.02409966   6.6399406   532 0.040249756
```

3기타패션(Fashion)의 연관규칙분석

rule_fashion<-subset(pm3_rule, items %pin% c("3"))

inspect(sort(rule_fashion,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                          rhs                                          support      confidence lift       count
[1]  {3기타패션-패션잡화-여성샌들}                 => {1의류-여성의류-여성원피스}                0.0005164034 0.04069389  0.8427998 1126
[2]  {1의류-여성의류-여성원피스}                   => {3기타패션-패션잡화-여성샌들}              0.0005164034 0.01069509  0.8427998 1126
[3]  {3기타패션-패션잡화-유아동샌들}               => {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} 0.0003801940 0.04689179  4.4024094  829
[4]  {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} => {3기타패션-패션잡화-유아동샌들}            0.0003801940 0.03569429  4.4024094  829
[5]  {3기타패션-패션잡화-여성샌들}                 => {1의류-여성의류-여성남방셔츠}             0.0003256185 0.02565956  0.6976893  710
[6]  {3기타패션-패션잡화-여성샌들}                 => {1의류-여성의류-여성티셔츠/탑}            0.0003205737 0.02526202  0.6737895  699
[7]  {3기타패션-패션잡화-야구모자}                 => {1의류-남성의류-남성티셔츠}                0.0002811326 0.09573637  1.3874113  613
[8]  {2스포츠-시즌스포츠-아쿠아슈즈}               => {3기타패션-패션잡화-유아동샌들}           0.0002788395 0.04723798  5.8261668  608
[9]  {3기타패션-패션잡화-유아동샌들}               => {2스포츠-시즌스포츠-아쿠아슈즈}           0.0002788395 0.03439109  5.8261668  608
[10] {3기타패션-패션잡화-여성샌들}                 => {1의류-여성의류-여성바지}                  0.0002783809 0.02193712  0.8052445  607
[11] {1의류-여성의류-여성바지}                     => {3기타패션-패션잡화-여성샌들}              0.0002783809 0.01021851  0.8052445  607
[12] {3기타패션-패션잡화-아동모}                   => {1의류-유아동의류-영유아티셔츠/탑}         0.0002673740 0.04323002  3.1962830  583
[13] {1의류-유아동의류-영유아티셔츠/탑}            => {3기타패션-패션잡화-아동모}                0.0002673740 0.01976874  3.1962830  583
[14] {1의류-속옷/양말/홈웨어-유아동일반양말}       => {3기타패션-패션잡화-아동모}                0.0002600362 0.06236252 10.0830007  567
[15] {3기타패션-패션잡화-아동모}                   => {1의류-속옷/양말/홈웨어-유아동일반양말}    0.0002600362 0.04204360 10.0830007  567
[16] {3기타패션-패션잡화-여성샌들}                 => {1의류-남성의류-남성티셔츠}                0.0002554500 0.02013010  0.2917254  557
[17] {3기타패션-패션잡화-기타패션잡화}             => {1의류-남성의류-남성티셔츠}                0.0002526983 0.06478542  0.9388702  551
[18] {3기타패션-패션잡화-여성슬리퍼}               => {3기타패션-패션잡화-여성샌들}              0.0002407742 0.05608974  4.4200137  525
[19] {3기타패션-패션잡화-여성샌들}                 => {3기타패션-패션잡화-여성슬리퍼}            0.0002407742 0.01897362  4.4200137  525
[20] {3기타패션-패션잡화-유아동샌들}               => {2스포츠-시즌스포츠-아동수영복}            0.0002283915 0.02816901  2.7824044  498
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_fashion_df<-as(rule_fashion,"data.frame")

rule_fashion_df<-transform(rule_fashion_df,IS=sqrt(lift*support))

rule_fashion_df[order(-rule_fashion_df$IS),]

```
                                                                                  rules      support confidence       lift count          IS
3                          {3기타패션-패션잡화-3단우산} => {3기타패션-패션잡화-장우산} 0.0001008959 0.21174206 80.5612218   220 0.090157052
4                          {3기타패션-패션잡화-장우산} => {3기타패션-패션잡화-3단우산} 0.0001008959 0.03838772 80.5612218   220 0.090157052
108                         {3기타패션-패션잡화-목걸이} => {3기타패션-패션잡화-귀걸이} 0.0002173847 0.12509897 30.5629190   474 0.081510194
109                         {3기타패션-패션잡화-귀걸이} => {3기타패션-패션잡화-목걸이} 0.0002173847 0.05310924 30.5629190   474 0.081510194
98            {3기타패션-패션잡화-유아동레인부츠/슈즈} => {3기타패션-패션잡화-아동우산} 0.0001238267 0.07812500 42.8874386   270 0.072873944
99            {3기타패션-패션잡화-아동우산} => {3기타패션-패션잡화-유아동레인부츠/슈즈} 0.0001238267 0.06797583 42.8874386   270 0.072873944
106                         {3기타패션-패션잡화-목걸이} => {3기타패션-패션잡화-팔찌} 0.0001073165 0.06175772 26.7502201   234 0.053579289
107                         {3기타패션-패션잡화-팔찌} => {3기타패션-패션잡화-목걸이} 0.0001073165 0.04648391 26.7502201   234 0.053579289
81         {4뷰티건강-건강식품-오메가3/기타추출오일} => {4뷰티건강-건강식품-일반비타민} 0.0001274957 0.08761425 21.1047153   278 0.051872537
82         {4뷰티건강-건강식품-일반비타민} => {4뷰티건강-건강식품-오메가3/기타추출오일} 0.0001274957 0.03071144 21.1047153   278 0.051872537
696              {1의류-속옷/양말/홈웨어-유아동일반양말} => {3기타패션-패션잡화-아동모} 0.0002600362 0.06236252 10.0830007   567 0.051204929
697              {3기타패션-패션잡화-아동모} => {1의류-속옷/양말/홈웨어-유아동일반양말} 0.0002600362 0.04204360 10.0830007   567 0.051204929
137                        {3기타패션-패션잡화-아동우산} => {3기타패션-패션잡화-장우산} 0.0001050234 0.05765358 21.9353795   229 0.047997173
138                        {3기타패션-패션잡화-장우산} => {3기타패션-패션잡화-아동우산} 0.0001050234 0.03995812 21.9353795   229 0.047997173
922      {3기타패션-패션잡화-유아동샌들} => {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} 0.0003801940 0.04689179  4.4024094   829 0.040911728
923      {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} => {3기타패션-패션잡화-유아동샌들} 0.0003801940 0.03569429  4.4024094   829 0.040911728
428                 {3기타패션-패션잡화-여성토트백} => {3기타패션-패션잡화-여성숄더백} 0.0002183020 0.05425738  7.6445059   476 0.040851079
429                 {3기타패션-패션잡화-여성숄더백} => {3기타패션-패션잡화-여성토트백} 0.0002183020 0.03075730  7.6445059   476 0.040851079
737                 {2스포츠-시즌스포츠-아쿠아슈즈} => {3기타패션-패션잡화-유아동샌들} 0.0002788395 0.04723798  5.8261668   608 0.040305897
738                 {3기타패션-패션잡화-유아동샌들} => {2스포츠-시즌스포츠-아쿠아슈즈} 0.0002788395 0.03439109  5.8261668   608 0.040305897
173                         {3기타패션-패션잡화-귀걸이} => {3기타패션-패션잡화-팔찌} 0.0001128199 0.02756303 11.9388636   246 0.036700703
172                         {3기타패션-패션잡화-팔찌} => {3기타패션-패션잡화-귀걸이} 0.0001128199 0.04886770 11.9388636   246 0.036700703
625    {1의류-스포츠패션-유아동런닝/트레이닝화} => {3기타패션-패션잡화-유아동스니커즈} 0.0002027090 0.03539965  6.0854406   442 0.035122263
626    {3기타패션-패션잡화-유아동스니커즈} => {1의류-스포츠패션-유아동런닝/트레이닝화} 0.0002027090 0.03484705  6.0854406   442 0.035122263
503                   {3기타패션-패션잡화-여성슬리퍼} => {3기타패션-패션잡화-여성샌들} 0.0002407742 0.05608974  4.4200137   525 0.032622467
504                   {3기타패션-패션잡화-여성샌들} => {3기타패션-패션잡화-여성슬리퍼} 0.0002407742 0.01897362  4.4200137   525 0.032622467
821                 {1의류-유아동의류-영유아티셔츠/탑} => {3기타패션-패션잡화-아동모} 0.0002673740 0.01976874  3.1962830   583 0.029233595
820                 {3기타패션-패션잡화-아동모} => {1의류-유아동의류-영유아티셔츠/탑} 0.0002673740 0.04323002  3.1962830   583 0.029233595
402          {3기타패션-패션잡화-유아동슬립온} => {3기타패션-패션잡화-유아동스니커즈} 0.0001229095 0.03581451  6.1567587   268 0.027508620
403          {3기타패션-패션잡화-유아동스니커즈} => {3기타패션-패션잡화-유아동슬립온} 0.0001229095 0.02112898  6.1567587   268 0.027508620
446              {3기타패션-패션잡화-여성크로스백} => {3기타패션-패션잡화-여성숄더백} 0.0001472162 0.03579393  5.0431285   321 0.027247576
447              {3기타패션-패션잡화-여성숄더백} => {3기타패션-패션잡화-여성크로스백} 0.0001472162 0.02074179  5.0431285   321 0.027247576
812                         {3기타패션-패션잡화-아동모} => {2스포츠-시즌스포츠-아동수영복} 0.0002123399 0.03433190  3.3911456   463 0.026834225
813                         {2스포츠-시즌스포츠-아동수영복} => {3기타패션-패션잡화-아동모} 0.0002123399 0.02097395  3.3911456   463 0.026834225
924                 {3기타패션-패션잡화-유아동샌들} => {2스포츠-시즌스포츠-아동수영복} 0.0002283915 0.02816901  2.7824044   498 0.025208682
925                 {2스포츠-시즌스포츠-아동수영복} => {3기타패션-패션잡화-유아동샌들} 0.0002283915 0.02255946  2.7824044   498 0.025208682
459                   {3기타패션-패션잡화-여성플랫} => {3기타패션-패션잡화-여성펌프스} 0.0001100682 0.02819218  5.2472969   240 0.024032490
460                   {3기타패션-패션잡화-여성펌프스} => {3기타패션-패션잡화-여성플랫} 0.0001100682 0.02048656  5.2472969   240 0.024032490
462                   {3기타패션-패션잡화-여성샌들} => {3기타패션-패션잡화-여성플랫} 0.0001605161 0.01264908  3.2398550   350 0.022804584
461                   {3기타패션-패션잡화-여성플랫} => {3기타패션-패션잡화-여성샌들} 0.0001605161 0.04111359  3.2398550   350 0.022804584
659  {3기타패션-패션잡화-유아동스니커즈} => {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} 0.0001774850 0.03051088  2.8644967   387 0.022547842
660  {1의류-스포츠패션-유아동스포츠샌들/슬리퍼} => {3기타패션-패션잡화-유아동스니커즈} 0.0001774850 0.01666308  2.8644967   387 0.022547842
1171                        {3기타패션-패션잡화-여성샌들} => {1의류-여성의류-여성원피스} 0.0005164034 0.04069389  0.8427998  1126 0.020862039
1172                        {1의류-여성의류-여성원피스} => {3기타패션-패션잡화-여성샌들} 0.0005164034 0.01069509  0.8427998  1126 0.020862039
392                         {3기타패션-패션잡화-야구모자} => {1의류-남성의류-남성티셔츠} 0.0002811326 0.09573637  1.3874113   613 0.019749595
616                   {3기타패션-패션잡화-여성펌프스} => {3기타패션-패션잡화-여성샌들} 0.0001504266 0.02799829  2.2063363   328 0.018217892
617                   {3기타패션-패션잡화-여성샌들} => {3기타패션-패션잡화-여성펌프스} 0.0001504266 0.01185399  2.2063363   328 0.018217892
810                         {3기타패션-패션잡화-아동모} => {1의류-유아동의류-영유아바지} 0.0001096096 0.01772208  2.7659006   239 0.017411756
811                         {1의류-유아동의류-영유아바지} => {3기타패션-패션잡화-아동모} 0.0001096096 0.01710686  2.7659006   239 0.017411756
818                         {3기타패션-패션잡화-아동모} => {1의류-유아동의류-남아바지} 0.0001449231 0.02343171  2.0890559   316 0.017399786
819                         {1의류-유아동의류-남아바지} => {3기타패션-패션잡화-아동모} 0.0001449231 0.01292064  2.0890559   316 0.017399786
```

4뷰티/건강(BH)의 연관규칙분석

rule_bh<-subset(pm3_rule, items %pin% c("4"))

inspect(sort(rule_bh,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                                    rhs                                                    support      confidence lift      count
[1]  {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} => {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   0.0016867954 0.08046379 3.431690 3678
[2]  {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   => {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} 0.0016867954 0.07193991 3.431690 3678
[3]  {4뷰티건강-화장품/뷰티케어-스킨/토너}                => {4뷰티건강-화장품/뷰티케어-크림/밤/오일}                0.0009979518 0.11614625 10.242374 2176
[4]  {4뷰티건강-화장품/뷰티케어-크림/밤/오일}             => {4뷰티건강-화장품/뷰티케어-스킨/토너}                  0.0009979518 0.08800453 10.242374 2176
[5]  {4뷰티건강-화장품/뷰티케어-메이크업베이스/프라이머} => {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   0.0009575935 0.13365766 5.700348 2088
[6]  {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   => {4뷰티건강-화장품/뷰티케어-메이크업베이스/프라이머} 0.0009575935 0.04084028 5.700348 2088
[7]  {4뷰티건강-화장품/뷰티케어-에센스/세럼}              => {4뷰티건강-화장품/뷰티케어-크림/밤/오일}                0.0009539245 0.12290239 10.838165 2080
[8]  {4뷰티건강-화장품/뷰티케어-크림/밤/오일}             => {4뷰티건강-화장품/뷰티케어-에센스/세럼}                0.0009539245 0.08412198 10.838165 2080
[9]  {4뷰티건강-화장품/뷰티케어-립글로즈/틴트}            => {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} 0.0007947842 0.12335398 5.884252 1733
[10] {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} => {4뷰티건강-화장품/뷰티케어-립글로즈/틴트}            0.0007947842 0.03791293 5.884252 1733
[11] {4뷰티건강-화장품/뷰티케어-메이크업세트}             => {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   0.0007833188 0.06695939 2.855742 1708
[12] {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   => {4뷰티건강-화장품/뷰티케어-메이크업세트}             0.0007833188 0.03340766 2.855742 1708
[13] {4뷰티건강-화장품/뷰티케어-기름종이}                 => {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   0.0007691017 0.15545050 6.629788 1677
[14] {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   => {4뷰티건강-화장품/뷰티케어-기름종이}                 0.0007691017 0.03280131 6.629788 1677
[15] {4뷰티건강-화장품/뷰티케어-아이섀도우}               => {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} 0.0007571776 0.13017425 6.209594 1651
[16] {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} => {4뷰티건강-화장품/뷰티케어-아이섀도우}               0.0007571776 0.03611901 6.209594 1651
[17] {4뷰티건강-화장품/뷰티케어-메이크업세트}             => {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} 0.0007525914 0.06433276 3.068812 1641
[18] {4뷰티건강-화장품/뷰티케어-볼륨서/쉐이빙/하이라이터} => {4뷰티건강-화장품/뷰티케어-메이크업세트}             0.0007525914 0.03590024 3.068812 1641
[19] {4뷰티건강-화장품/뷰티케어-스킨케어세트}             => {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   0.0007227813 0.03668614 1.564622 1576
[20] {4뷰티건강-화장품/뷰티케어-BB/파운데이션/컬쳐트튜}   => {4뷰티건강-화장품/뷰티케어-스킨케어세트}             0.0007227813 0.03082580 1.564622 1576
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_bh_df<-as(rule_bh,"data.frame")

rule_bh_df<-transform(rule_bh_df,IS=sqrt(lift*support))

rule_bh_df[order(-rule_bh_df$IS),]

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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5출산육아(Baby)의 연관규칙분석

rule_baby<-subset(pm3_rule, items %pin% c("5"))

inspect(sort(rule_baby,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                                   rhs                                                  support      confidence lift      count
[1]  {5출산육아-출산/육아용품-유아용샴푸/바디워시} => {5출산육아-출산/육아용품-유아용화장품}        0.0006494025 0.28542633 30.878809 1416
[2]  {5출산육아-출산/육아용품-유아용화장품}         => {5출산육아-출산/육아용품-유아용샴푸/바디워시} 0.0006494025 0.07025552 30.878809 1416
[3]  {5출산육아-출산/육아용품-유아목욕용품}         => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0002389397 0.08805138 23.393814  521
[4]  {5출산육아-출산/육아용품-출산/신생아용품세트} => {5출산육아-출산/육아용품-유아목욕용품}        0.0002389397 0.06348239 23.393814  521
[5]  {5출산육아-출산/육아용품-유아용세척용품}       => {6홈케어-세제/위생-액상세탁세제}                0.0002306846 0.15553494 28.730824  503
[6]  {6홈케어-세제/위생-액상세탁세제}               => {5출산육아-출산/육아용품-유아용세척용품}       0.0002306846 0.04261267 28.730824  503
[7]  {5출산육아-출산/육아용품-젖병/젖꼭지}          => {5출산육아-출산/육아용품-이유식용품}           0.0002192192 0.09751122 20.680858  478
[8]  {5출산육아-출산/육아용품-이유식용품}           => {5출산육아-출산/육아용품-젖병/젖꼭지}          0.0002192192 0.04649353 20.680858  478
[9]  {5출산육아-출산/육아용품-유아용기저귀}         => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001875746 0.05617360 14.924410  409
[10] {5출산육아-출산/육아용품-출산/신생아용품세트} => {5출산육아-출산/육아용품-유아용기저귀}         0.0001875746 0.04983551 14.924410  409
[11] {5출산육아-출산/육아용품-유아용기저귀}         => {5출산육아-출산/육아용품-유아용물티슈}         0.0001715230 0.05136657 10.067691  374
[12] {5출산육아-출산/육아용품-유아용물티슈}         => {5출산육아-출산/육아용품-유아용기저귀}         0.0001715230 0.03361798 10.067691  374
[13] {5출산육아-출산/육아용품-유아목욕용품}         => {6홈케어-세제/위생-액상세탁세제}                0.0001701471 0.06270069 11.582237  371
[14] {6홈케어-세제/위생-액상세탁세제}               => {5출산육아-출산/육아용품-유아목욕용품}         0.0001701471 0.03143002 11.582237  371
[15] {5출산육아-출산/육아용품-유아목욕용품}         => {5출산육아-출산/육아용품-유아용화장품}         0.0001605161 0.05915160  6.399308  350
[16] {5출산육아-출산/육아용품-유아용화장품}         => {5출산육아-출산/육아용품-유아목욕용품}         0.0001605161 0.01736542  6.399308  350
[17] {5출산육아-출산/육아용품-유아용세척용품}       => {5출산육아-출산/육아용품-이유식용품}           0.0001595989 0.10760668 22.821973  348
[18] {5출산육아-출산/육아용품-이유식용품}           => {5출산육아-출산/육아용품-유아용세척용품}       0.0001595989 0.03384885 22.821973  348
[19] {5출산육아-출산/육아용품-출산/신생아용품세트} => {1의류-속옷/양말/홈웨어-유아동내의}           0.0001554714 0.04130620  4.232660  339
[20] {1의류-속옷/양말/홈웨어-유아동내의}           => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001554714 0.01593120  4.232660  339
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_baby_df<-as(rule_baby,"data.frame")

rule_baby_df<-transform(rule_baby_df,IS=sqrt(lift*support))

rule_baby_df[order(-rule_baby_df$IS),]

```
                                                                                  rules      support confidence       lift count          IS
295  {5출산육아-출산/육아용품-유아용샴푸/바디워시} => {5출산육아-출산/육아용품-유아용화장품} 0.0006494025 0.28542633 30.8788091  1416 0.141607821
296  {5출산육아-출산/육아용품-유아용화장품} => {5출산육아-출산/육아용품-유아용샴푸/바디워시} 0.0006494025 0.07025552 30.8788091  1416 0.141607821
147              {5출산육아-출산/육아용품-유아용세척용품} => {6홈케어-세제/위생-액상세탁세제} 0.0002306846 0.15553494 28.7308244   503 0.081411054
148              {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아용세척용품} 0.0002306846 0.04261267 28.7308244   503 0.081411054
383  {5출산육아-출산/육아용품-유아목욕용품} => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0002389397 0.08805138 23.3938144   521 0.074764378
384  {5출산육아-출산/육아용품-출산/신생아용품세트} => {5출산육아-출산/육아용품-유아목욕용품} 0.0002389397 0.06348239 23.3938144   521 0.074764378
141        {5출산육아-출산/육아용품-유아용세척용품} => {5출산육아-출산/육아용품-유아목욕용품} 0.0001490507 0.10049474 37.0331876   325 0.074295510
142        {5출산육아-출산/육아용품-유아목욕용품} => {5출산육아-출산/육아용품-유아용세척용품} 0.0001490507 0.05492648 37.0331876   325 0.074295510
139         {5출산육아-출산/육아용품-유아용세척용품} => {5출산육아-출산/육아용품-젖병/젖꼭지} 0.0001352922 0.09121831 40.5749518   295 0.074090983
140         {5출산육아-출산/육아용품-젖병/젖꼭지} => {5출산육아-출산/육아용품-유아용세척용품} 0.0001352922 0.06017952 40.5749518   295 0.074090983
174              {5출산육아-출산/육아용품-젖병/젖꼭지} => {5출산육아-출산/육아용품-이유식용품} 0.0002192192 0.09751122 20.6808579   478 0.067332318
175              {5출산육아-출산/육아용품-이유식용품} => {5출산육아-출산/육아용품-젖병/젖꼭지} 0.0002192192 0.04649353 20.6808579   478 0.067332318
143          {5출산육아-출산/육아용품-유아용세척용품} => {5출산육아-출산/육아용품-이유식용품} 0.0001595989 0.10760668 22.8219731   348 0.060351985
144          {5출산육아-출산/육아용품-이유식용품} => {5출산육아-출산/육아용품-유아용세척용품} 0.0001595989 0.03384885 22.8219731   348 0.060351985
338  {5출산육아-출산/육아용품-출산/신생아용품세트} => {5출산육아-출산/육아용품-유아용기저귀} 0.0001875746 0.04983551 14.9244097   409 0.052909734
337  {5출산육아-출산/육아용품-유아용기저귀} => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001875746 0.05617360 14.9244097   409 0.052909734
387                 {5출산육아-출산/육아용품-유아목욕용품} => {6홈케어-세제/위생-액상세탁세제} 0.0001701471 0.06270069 11.5822373   371 0.044392390
388                 {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아목욕용품} 0.0001701471 0.03143002 11.5822373   371 0.044392390
339          {5출산육아-출산/육아용품-유아용기저귀} => {5출산육아-출산/육아용품-유아용물티슈} 0.0001715230 0.05136657 10.0676908   374 0.041555267
340          {5출산육아-출산/육아용품-유아용물티슈} => {5출산육아-출산/육아용품-유아용기저귀} 0.0001715230 0.03361798 10.0676908   374 0.041555267
145    {5출산육아-출산/육아용품-유아용세척용품} => {6홈케어-세제/위생-섬유유연제/향기지속제} 0.0001008959 0.06802721 14.3509114   220 0.038051907
146    {6홈케어-세제/위생-섬유유연제/향기지속제} => {5출산육아-출산/육아용품-유아용세척용품} 0.0001008959 0.02128483 14.3509114   220 0.038051907
389          {5출산육아-출산/육아용품-유아목욕용품} => {5출산육아-출산/육아용품-유아용화장품} 0.0001605161 0.05915160  6.3993077   350 0.032049839
390          {5출산육아-출산/육아용품-유아용화장품} => {5출산육아-출산/육아용품-유아목욕용품} 0.0001605161 0.01736542  6.3993077   350 0.032049839
510 {5출산육아-출산/육아용품-출산/신생아용품세트} => {1의류-속옷/양말/홈웨어-유아동일반양말} 0.0001215337 0.03228951  7.7437502   265 0.030677781
511 {1의류-속옷/양말/홈웨어-유아동일반양말} => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001215337 0.02914650  7.7437502   265 0.030677781
386              {5출산육아-출산/육아용품-이유식용품} => {5출산육아-출산/육아용품-유아목욕용품} 0.0001068579 0.02266317  8.3515736   233 0.029873593
385              {5출산육아-출산/육아용품-유아목욕용품} => {5출산육아-출산/육아용품-이유식용품} 0.0001068579 0.03937806  8.3515736   233 0.029873593
509    {5출산육아-출산/육아용품-이유식용품} => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001229095 0.02606750  6.9257103   268 0.029175943
508    {5출산육아-출산/육아용품-출산/신생아용품세트} => {5출산육아-출산/육아용품-이유식용품} 0.0001229095 0.03265505  6.9257103   268 0.029175943
512     {5출산육아-출산/육아용품-출산/신생아용품세트} => {1의류-속옷/양말/홈웨어-유아동내의} 0.0001554714 0.04130620  4.2326599   339 0.025652629
513     {1의류-속옷/양말/홈웨어-유아동내의} => {5출산육아-출산/육아용품-출산/신생아용품세트} 0.0001554714 0.01593120  4.2326599   339 0.025652629
748                 {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아용화장품} 0.0001302474 0.02405964  2.6028890   284 0.018412482
749                 {5출산육아-출산/육아용품-유아용화장품} => {6홈케어-세제/위생-액상세탁세제} 0.0001302474 0.01409080  2.6028890   284 0.018412482
975             {4뷰티건강-화장품/뷰티케어-선크림류} => {5출산육아-출산/육아용품-유아용화장품} 0.0001449231 0.01671427  1.8082311   316 0.016188099
974             {5출산육아-출산/육아용품-유아용화장품} => {4뷰티건강-화장품/뷰티케어-선크림류} 0.0001449231 0.01567849  1.8082311   316 0.016188099
623          {5출산육아-출산/육아용품-유아용물티슈} => {5출산육아-출산/육아용품-유아용화장품} 0.0001109854 0.02175281  2.3533248   242 0.016161213
624          {5출산육아-출산/육아용품-유아용화장품} => {5출산육아-출산/육아용품-유아용물티슈} 0.0001109854 0.01200695  2.3533248   242 0.016161213
976                       {5출산육아-출산/육아용품-유아용화장품} => {1의류-남성의류-남성티셔츠} 0.0001137372 0.01230464  0.1783188   248 0.004503496
```

6홈케어(House)의 연관규칙분석

rule_house<-subset(pm3_rule, items %pin% c("6"))

inspect(sort(rule_house,by="support")[1:20]) #지지도 순위로 정렬

```
     lhs                                          rhs                                          support      confidence lift      count
[1]  {6홈케어-세제/위생-섬유유연제/향기지속제}  => {6홈케어-세제/위생-액상세탁세제}           0.0011052683 0.23316563  43.07097 2410
[2]  {6홈케어-세제/위생-액상세탁세제}           => {6홈케어-세제/위생-섬유유연제/향기지속제} 0.0011052683 0.20416808  43.07097 2410
[3]  {6홈케어-식기/조리기구-대접/볼}            => {6홈케어-식기/조리기구-접시}               0.0002802153 0.22646405  77.97208  611
[4]  {6홈케어-식기/조리기구-접시}               => {6홈케어-식기/조리기구-대접/볼}            0.0002802153 0.09647876  77.97208  611
[5]  {5출산육아-출산/육아용품-유아용세척용품}   => {6홈케어-세제/위생-액상세탁세제}           0.0002306846 0.15553494  28.73082  503
[6]  {6홈케어-세제/위생-액상세탁세제}           => {5출산육아-출산/육아용품-유아용세척용품}   0.0002306846 0.04261267  28.73082  503
[7]  {6홈케어-식기/조리기구-냄비}               => {6홈케어-식기/조리기구-프라이팬}           0.0002279329 0.10827887  29.92754  497
[8]  {6홈케어-식기/조리기구-프라이팬}           => {6홈케어-식기/조리기구-냄비}               0.0002279329 0.06299911  29.92754  497
[9]  {6홈케어-식기/조리기구-찬기/종지}           => {6홈케어-식기/조리기구-접시}               0.0001985814 0.38217123 131.58240  433
[10] {6홈케어-식기/조리기구-접시}               => {6홈케어-식기/조리기구-찬기/종지}           0.0001985814 0.06837202 131.58240  433
[11] {6홈케어-식기/조리기구-밥공기}             => {6홈케어-식기/조리기구-접시}               0.0001724402 0.24211204  83.35971  376
[12] {6홈케어-식기/조리기구-접시}               => {6홈케어-식기/조리기구-밥공기}             0.0001724402 0.05937155  83.35971  376
[13] {5출산육아-출산/육아용품-유아목욕용품}     => {6홈케어-세제/위생-액상세탁세제}           0.0001701471 0.06270069  11.58224  371
[14] {6홈케어-세제/위생-액상세탁세제}           => {5출산육아-출산/육아용품-유아목욕용품}     0.0001701471 0.03143002  11.58224  371
[15] {6홈케어-세제/위생-각티슈/미용티슈}         => {6홈케어-세제/위생-롤티슈}                 0.0001499679 0.18237591  83.17600  327
[16] {6홈케어-세제/위생-롤티슈}                 => {6홈케어-세제/위생-각티슈/미용티슈}         0.0001499679 0.06839573  83.17600  327
[17] {6홈케어-주방잡화-기타주방정리용품/소모품} => {6홈케어-식기/조리기구-반찬통/밀폐용기}     0.0001453818 0.05071189  15.50414  317
[18] {6홈케어-식기/조리기구-반찬통/밀폐용기}     => {6홈케어-주방잡화-기타주방정리용품/소모품} 0.0001453818 0.04444756  15.50414  317
[19] {6홈케어-식기/조리기구-커피잔}             => {6홈케어-식기/조리기구-접시}               0.0001426301 0.10045220  34.58591  311
[20] {6홈케어-식기/조리기구-접시}               => {6홈케어-식기/조리기구-커피잔}             0.0001426301 0.04910785  34.58591  311
```

#IS measure 순으로 연관규칙 선호지수 살펴 봄.

rule_house_df<-as(rule_house,"data.frame")

rule_house_df<-transform(rule_house_df,IS=sqrt(lift*support))

rule_house_df[order(-rule_house_df$IS),]

```
                                                                                  rules      support confidence       lift count         IS
644          {6홈케어-세제/위생-섬유유연제/향기지속제} => {6홈케어-세제/위생-액상세탁세제} 0.0011052683 0.23316563  43.070971  2410 0.21818565
645          {6홈케어-세제/위생-액상세탁세제} => {6홈케어-세제/위생-섬유유연제/향기지속제} 0.0011052683 0.20416808  43.070971  2410 0.21818565
19                       {6홈케어-식기/조리기구-찬기/종지} => {6홈케어-식기/조리기구-접시} 0.0001985814 0.38217123 131.582404   433 0.16164720
20                       {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-찬기/종지} 0.0001985814 0.06837202 131.582404   433 0.16164720
96                         {6홈케어-식기/조리기구-대접/볼} => {6홈케어-식기/조리기구-접시} 0.0002802153 0.22646405  77.972076   611 0.14781397
97                         {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-대접/볼} 0.0002802153 0.09647876  77.972076   611 0.14781397
27                       {6홈케어-식기/조리기구-밥공기} => {6홈케어-식기/조리기구-대접/볼} 0.0001224509 0.17192531 138.946362   267 0.13043813
28                       {6홈케어-식기/조리기구-대접/볼} => {6홈케어-식기/조리기구-밥공기} 0.0001224509 0.09896219 138.946362   267 0.13043813
29                         {6홈케어-식기/조리기구-밥공기} => {6홈케어-식기/조리기구-접시} 0.0001724402 0.24211204  83.359715   376 0.11989398
30                         {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-밥공기} 0.0001724402 0.05937155  83.359715   376 0.11989398
33                       {6홈케어-세제/위생-각티슈/미용티슈} => {6홈케어-세제/위생-롤티슈} 0.0001499679 0.18237591  83.176001   327 0.11168587
34                       {6홈케어-세제/위생-롤티슈} => {6홈케어-세제/위생-각티슈/미용티슈} 0.0001499679 0.06839573  83.176001   327 0.11168587
14                 {6홈케어-주방잡화-국자/뒤지개/주걱} => {6홈케어-주방잡화-기타조리도구} 0.0001018131 0.21915104 112.780596   222 0.10715662
15                 {6홈케어-주방잡화-기타조리도구} => {6홈케어-주방잡화-국자/뒤지개/주걱} 0.0001018131 0.05239556 112.780596   222 0.10715662
21                           {6홈케어-세제/위생-키친타올} => {6홈케어-세제/위생-롤티슈} 0.0001183233 0.19282511  87.941561   258 0.10200754
22                           {6홈케어-세제/위생-롤티슈} => {6홈케어-세제/위생-키친타올} 0.0001183233 0.05396361  87.941561   258 0.10200754
157                      {6홈케어-식기/조리기구-프라이팬} => {6홈케어-식기/조리기구-냄비} 0.0002279329 0.06299911  29.927543   497 0.08259221
156                      {6홈케어-식기/조리기구-냄비} => {6홈케어-식기/조리기구-프라이팬} 0.0002279329 0.10827887  29.927543   497 0.08259221
147          {5출산육아-출산/육아용품-유아용세척용품} => {6홈케어-세제/위생-액상세탁세제} 0.0002306846 0.15553494  28.730824   503 0.08141105
148          {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아용세척용품} 0.0002306846 0.04261267  28.730824   503 0.08141105
85                         {6홈케어-식기/조리기구-커피잔} => {6홈케어-식기/조리기구-접시} 0.0001426301 0.10045220  34.585915   311 0.07023526
86                         {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-커피잔} 0.0001426301 0.04910785  34.585915   311 0.07023526
40                       {6홈케어-식기/조리기구-음료용컵} => {6홈케어-식기/조리기구-접시} 0.0001041062 0.10992736  37.848235   227 0.06277129
41                       {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-음료용컵} 0.0001041062 0.03584399  37.848235   227 0.06277129
216          {6홈케어-주방잡화-기타주방정리용품/소모품} => {6홈케어-주방잡화-기타조리도구} 0.0001371267 0.04783235  24.615720   299 0.05809881
215          {6홈케어-주방잡화-기타조리도구} => {6홈케어-주방잡화-기타주방정리용품/소모품} 0.0001371267 0.07056880  24.615720   299 0.05809881
55                 {6홈케어-식기/조리기구-반상기세트/홈세트} => {6홈케어-식기/조리기구-접시} 0.0001036476 0.09164639  31.554056   226 0.05718830
56                 {6홈케어-식기/조리기구-접시} => {6홈케어-식기/조리기구-반상기세트/홈세트} 0.0001036476 0.03568609  31.554056   226 0.05718830
102          {6홈케어-주방잡화-주방선반/걸이대} => {6홈케어-식기/조리기구-반찬통/밀폐용기} 0.0001210750 0.07669959  23.449363   264 0.05328351
103          {6홈케어-식기/조리기구-반찬통/밀폐용기} => {6홈케어-주방잡화-주방선반/걸이대} 0.0001210750 0.03701626  23.449363   264 0.05328351
100        {6홈케어-주방잡화-주방선반/걸이대} => {6홈케어-주방잡화-기타주방정리용품/소모품} 0.0001054820 0.06682162  23.308632   230 0.04958470
101        {6홈케어-주방잡화-기타주방정리용품/소모품} => {6홈케어-주방잡화-주방선반/걸이대} 0.0001054820 0.03679411  23.308632   230 0.04958470
381 {6홈케어-주방잡화-기타주방정리용품/소모품} => {6홈케어-식기/조리기구-반찬통/밀폐용기} 0.0001453818 0.05071189  15.504142   317 0.04747652
382 {6홈케어-식기/조리기구-반찬통/밀폐용기} => {6홈케어-주방잡화-기타주방정리용품/소모품} 0.0001453818 0.04444756  15.504142   317 0.04747652
387                {5출산육아-출산/육아용품-유아목욕용품} => {6홈케어-세제/위생-액상세탁세제} 0.0001701471 0.06270069  11.582237   371 0.04439239
388                {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아목욕용품} 0.0001701471 0.03143002  11.582237   371 0.04439239
145 {5출산육아-출산/육아용품-유아용세척용품} => {6홈케어-세제/위생-섬유유연제/향기지속제} 0.0001008959 0.06802721  14.350911   220 0.03805191
146 {6홈케어-세제/위생-섬유유연제/향기지속제} => {5출산육아-출산/육아용품-유아용세척용품} 0.0001008959 0.02128483  14.350911   220 0.03805191
748                {6홈케어-세제/위생-액상세탁세제} => {5출산육아-출산/육아용품-유아용화장품} 0.0001302474 0.02405964   2.602889   284 0.01841248
749                {5출산육아-출산/육아용품-유아용화장품} => {6홈케어-세제/위생-액상세탁세제} 0.0001302474 0.01409080   2.602889   284 0.01841248
```

Ⅲ 수요 트렌드 예측 (순차패턴분석, 검색어 트렌드)

Ⅲ-(1) 순차패턴분석 (Sequence Pattern Analysis)

연관규칙과 동일한 상품분류를 이용하여 분석할 예정.

분류	상품 군
1의류	여성의류 스포츠패션 남성의류 유아동의류 속옷/양말/홈웨어
2스포츠	시즌스포츠 구기/필드스포츠 아웃도어/레저
3기타패션	패션잡화
4뷰티/건강	퍼스널케어, 화장품/뷰티케어 건강식품
5출산/육아	출산/육아용품
6홈케어	식기/조리기구 주방잡화 세제/위생

```
library(tidyverse)   library(arulesSequences)
setwd("C:/Users/USER/Desktop/lpoint")
library(arulesSequences)
product<-read_rds("product.rds")
master<-read_rds("master.rds")
session<-read_rds("session.rds")
```

지정한 상품군에 필요한 상품군들 뽑음

```
pm<-left_join(product,master)%>%filter(CLAC1_NM%in%c("건강식품","구기/필드스포츠","남성의류","세제/위생","속옷/양말/홈웨어","스포츠패션","시즌스포츠","식기/조리기구","아웃도어/레저","여성의류","유아동의류","주방잡화","출산/육아용품","패션잡화","퍼스널케어","화장품/뷰티케어"))
pms<-inner_join(pm, session)
pms_seq<-
pms%>%arrange(CLNT_ID,SESS_DT,SESS_SEQ,HITS_SEQ)%>%select(CLNT_ID,SESS_DT,CLAC1_NM,CLAC2_NM,CLAC3_NM)
```

지정한 상품군으로 묶어서 이름 다시 지음.

```
pms_seq<-pms_seq%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("여성의류" = "1의류-여성의류", "스포츠패션" = "1의류-스포츠패션", "남성의류" = "1의류-남성의류","유아동의류"="1의류-유아동의류","속옷/양말/홈웨어"="1의류-속옷/양말/홈웨어")))
pms_seq<-pms_seq%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("시즌스포츠" = "2스포츠-시즌스포츠", "구기/필드스포츠" = "2스포츠-구기/필드스포츠", "아웃도어/레저" = "2스포츠-아웃도어/레저")))
pms_seq<-pms_seq%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("건강식품" = "4뷰티건강-건강식품", "퍼스널케어" = "4뷰티건강-퍼스널케어", "화장품/뷰티케어" = "4뷰티건강-화장품/뷰티케어","출산/육아용품" = "5출산육아-출산/육아용품")))
pms_seq<-pms_seq%>%mutate(CLAC1_NM=stringr::str_replace_all(CLAC1_NM,c("패션잡화" = "3기타패션-패션잡화", "식기/조리기구" = "6홈케어-식기/조리기구", "주방잡화" = "6홈케어-주방잡화","세
```

제/위생"="6홈케어-세제/위생")))

pms_seq<-

pms_seq%>%filter(!is.na(CLAC3_NM))%>%unite(CLAC3_RNM,CLAC1_NM,CLAC3_NM,sep="-")%>%select(CLNT_ID,SESS_DT,CLAC3_RNM)

```
# A tibble: 3,987,262 x 3
   CLNT_ID SESS_DT    CLAC3_RNM
   <chr>   <date>     <chr>
 1 0000008 2018-06-05 3기타패션-패션잡화-여성클러치백
 2 0000008 2018-06-06 3기타패션-패션잡화-여성클러치백
 3 0000014 2018-09-11 3기타패션-패션잡화-여성시계
 4 0000014 2018-09-11 3기타패션-패션잡화-여성시계
 5 0000025 2018-07-03 4뷰티건강-화장품/뷰티케어-페이셜팩류
 6 0000033 2018-04-07 4뷰티건강-화장품/뷰티케어-페이셜클렌저
 7 0000033 2018-04-11 4뷰티건강-화장품/뷰티케어-스킨/토너
 8 0000033 2018-04-24 4뷰티건강-퍼스널케어-샴푸
 9 0000033 2018-05-08 6홈케어-세제/위생-섬유유연제/향기지속제
10 0000033 2018-05-08 4뷰티건강-퍼스널케어-샴푸
# ... with 3,987,252 more rows
```

CLNT_ID, SESS_DT별로 순서대로 산 항목들을 콤마로 이어 스트링으로 만들어 하나의 obs로 만듦.

pms_seq1<-

pms_seq%>%group_by(CLNT_ID,SESS_DT)%>%summarize(ITEMS=paste(CLAC3_RNM,collapse=","))

pms_seq1$SESS_DT<-as.character(pms_seq1$SESS_DT)

pms_seq1<-pms_seq1%>%separate(SESS_DT,c("YEAR","MONTH","DATE"),sep="-")%>%unite(SESS_DT,YEAR,MONTH,DATE,sep="")

pms_seq1$CLNT_ID<-as.integer(pms_seq1$CLNT_ID)

pms_seq1$SESS_DT<-as.integer(pms_seq1$SESS_DT)

스트링 안의 항목이 몇 개인지 콤마개수+1로 만들어 SIZE라는 변수만듦.

library(stringr)

pms_seq1<-pms_seq1%>%mutate(SIZE=str_count(ITEMS,","))%>%mutate(SIZE=SIZE+1)

pms_seq1<-pms_seq1%>%select(ITEMS,CLNT_ID,SESS_DT,SIZE)

변수가 다 factor이여야 transaction으로 불러올 때 좋다.

pms_seq_df<-as.data.frame(pms_seq1)

pms_seq_df$ITEMS<-as.factor(pms_seq_df$ITEMS)

pms_seq_df$CLNT_ID<-as.factor(pms_seq_df$CLNT_ID)

pms_seq_df$SESS_DT<-as.factor(pms_seq_df$SESS_DT)

pms_seq_df$SIZE<-as.factor(pms_seq_df$SIZE)

변수이름 재설정함.

pms_seq_df<-pms_seq_df%>%rename(items=ITEMS,sequenceID=CLNT_ID,eventID=SESS_DT)

pms_seq_df<-pms_seq_df%>%select(sequenceID,eventID,SIZE,items)

write.table(pms_seq_df,"pms_seq_df.csv",sep=",",row.names=F,col.names=F,quote=F)

pms_seq_trans<-read_baskets("pms_seq_df.csv",sep=",",info=c("sequenceID","eventID","SIZE"))

순차패턴을 한다. Min 지지도를 0.0001로 해둠. 패턴안에 2가지 항목이 있게 바꿈.

제일 높은 것부터 나열함.

```
pms_seq_rule<-cspade(pms_seq_trans,parameter=list(support=0.0001,
maxsize=5,maxlen=4),control=list(verbose=TRUE))
pms_seq_rule_df<-as(pms_seq_rule,"data.frame")
pms_seq_rule_size<-size(pms_seq_rule)
pms_seq_rule_dfsize<-cbind(pms_seq_rule_df,pms_seq_rule_size)
seqrule_subset2<-subset(pms_seq_rule_dfsize,subset=pms_seq_rule_size>=2)
```

지정한 상품군들 추려서 각각 살펴봄.

1CLOTHING

```
seqrule_clothing<-seqrule_subset2[grep("1",seqrule_subset2$sequence),]
suqrule_clothing_desc<-seqrule_clothing%>%arrange(desc(support))
suqrule_clothing_desc[1:300,]
```

	sequence	support	pms_seq_rule_size
1	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠}>	0.027220729	2
2	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스}>	0.021140038	2
3	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성티셔츠/탑}>	0.015173497	2
4	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성남방셔츠}>	0.014946423	2
5	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성캐주얼바지}>	0.012923617	2
6	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성원피스}>	0.011516982	2
7	<{1의류-남성의류-남성캐주얼바지},{1의류-남성의류-남성캐주얼바지}>	0.011465429	2
8	<{1의류-남성의류-남성캐주얼바지},{1의류-남성의류-남성티셔츠}>	0.011287452	2
9	<{1의류-여성의류-여성바지},{1의류-여성의류-여성바지}>	0.011255539	2
10	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성원피스}>	0.011183120	2
11	<{1의류-남성의류-남성티셔츠},{1의류-여성의류-여성원피스}>	0.010877490	2
12	<{1의류-남성의류-남성티셔츠},{1의류-여성의류-여성티셔츠/탑}>	0.010690921	2
13	<{1의류-여성의류-여성티셔츠/탑},{1의류-남성의류-남성티셔츠}>	0.010646733	2
14	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성남방셔츠}>	0.010398792	2
15	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성남방셔츠}>	0.010347240	2
16	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성티셔츠/탑}>	0.010314099	2
17	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠}>	0.010098072	3
18	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성티셔츠/탑}>	0.009718796	2
19	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성바지}>	0.009621829	2
20	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성바지}>	0.009405801	2
21	<{1의류-여성의류-여성원피스},{1의류-남성의류-남성티셔츠}>	0.009392299	2
22	<{1의류-여성의류-여성바지},{1의류-여성의류-여성티셔츠/탑}>	0.008860822	2
23	<{1의류-여성의류-여성바지},{1의류-여성의류-여성남방셔츠}>	0.008782266	2
24	<{1의류-여성의류-여성바지},{1의류-여성의류-여성원피스}>	0.008773674	2
25	<{1의류-남성의류-남성티셔츠},{1의류-여성의류-여성남방셔츠}>	0.008728259	2
26	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스}>	0.008621473	3
27	<{1의류-여성의류-여성남방셔츠},{1의류-남성의류-남성티셔츠}>	0.008508549	2
28	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성바지}>	0.008372304	2
29	<{1의류-남성의류-남성티셔츠},{1의류-여성의류-여성바지}>	0.007945158	2
30	<{1의류-여성의류-여성바지},{1의류-남성의류-남성티셔츠}>	0.006987763	2
31	<{1의류-여성의류-여성스커트},{1의류-여성의류-여성원피스}>	0.006745959	2
32	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성스커트}>	0.006445238	2
33	<{1의류-여성의류-여성티셔츠/탑},{1의류-남성의류-남성캐주얼바지}>	0.006227983	2
34	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성남방셔츠}>	0.006106467	3
35	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성티셔츠/탑}>	0.005957948	3
36	<{1의류-남성의류-남성캐주얼바지},{1의류-여성의류-여성원피스}>	0.005811884	2
37	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성스커트}>	0.005723509	2
38	<{1의류-여성의류-여성원피스},{1의류-남성의류-남성캐주얼바지}>	0.005654773	2
39	<{1의류-여성의류-여성스커트},{1의류-여성의류-여성스커트}>	0.005640044	2
40	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성캐주얼바지}>	0.005530802	3
41	<{1의류-여성의류-여성스커트},{1의류-여성의류-여성남방셔츠}>	0.005518528	2
42	<{1의류-스포츠패션-남성스포츠티셔츠},{1의류-스포츠패션-남성스포츠티셔츠}>	0.005457157	2
43	<{1의류-남성의류-남성캐주얼바지},{1의류-여성의류-여성티셔츠/탑}>	0.005443655	2
44	<{1의류-여성의류-여성남방셔츠},{1의류-남성의류-남성캐주얼바지}>	0.005016509	2
45	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스}>	0.005001780	3
46	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성스커트}>	0.004999325	2
47	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성캐주얼바지,1의류-남성의류-남성티셔츠}>	0.004876582	2
48	<{1의류-유아동의류-영유아티셔츠/탑},{1의류-유아동의류-영유아티셔츠/탑}>	0.004874127	2
49	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성캐주얼바지},{1의류-남성의류-남성티셔츠}>	0.004874127	3
50	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠}>	0.004839759	4
51	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스}>	0.004829939	3
52	<{1의류-여성의류-여성스커트},{1의류-여성의류-여성티셔츠/탑}>	0.004816438	2
53	<{1의류-남성의류-남성티셔츠},{1의류-스포츠패션-남성스포츠티셔츠}>	0.004809073	2

사실 결과가 더 많았고, 그 결과중 고른 규칙들도 있지만, 어떤형식으로 결과가 나오는지만 보여주기 위하여 부분만 캡처함

2스포츠

seqrule_sports<-seqrule_subset2[grep("2",seqrule_subset2$sequence),]

suqrule_sports_desc<-seqrule_sports%>%arrange(desc(support))

suqrule_sports_desc[1:300,]

```
                                                                                                  sequence      support pms_seq_rule_size
1                                                <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-아동수영복}> 0.0027789029                2
2                                            <{2스포츠-시즌스포츠-아동수영복},{2스포츠-시즌스포츠-아동수영복}> 0.0027789029                2
3                                                <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-남성수영복}> 0.0025776043                2
4                                            <{1의류-유아동의류-여아티셔츠/탑},{2스포츠-시즌스포츠-아동수영복}> 0.0023358005                2
5                                           <{1의류-유아동의류-영유아티셔츠/탑},{2스포츠-시즌스포츠-아동수영복}> 0.0023284359                2
6                                                <{1의류-유아동의류-남아바지},{2스포츠-시즌스포츠-아동수영복}> 0.0020903143                2
7                                             <{1의류-여성의류-여성티셔츠/탑},{2스포츠-시즌스포츠-아동수영복}> 0.0020547189                2
8                                              <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-여성비치웨어}> 0.0020240331                2
9                                                <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-아루아슈즈}> 0.0019368855                2
10                                               <{1의류-여성의류-여성원피스},{2스포츠-시즌스포츠-아동수영복}> 0.0019332032                2
11                                               <{2스포츠-시즌스포츠-아동수영복},{1의류-남성의류-남성티셔츠}> 0.0018681494                2
12                                     <{1의류-스포츠패션-유아동스포츠샌들/슬리퍼},{2스포츠-시즌스포츠-아동수영복}> 0.0018350088                2
13                                               <{2스포츠-시즌스포츠-아동수영복},{1의류-여성의류-여성원피스}> 0.0017711824                2
14                                             <{1의류-여성의류-여성원피스},{2스포츠-시즌스포츠-여성비치웨어}> 0.0017343595                2
15                                            <{1의류-여성의류-여성티셔츠/탑},{2스포츠-시즌스포츠-남성수영복}> 0.0017184029                2
16                                           <{1의류-여성의류-여성티셔츠/탑},{2스포츠-시즌스포츠-여성비치웨어}> 0.0017147206                2
17                                               <{1의류-여성의류-여성원피스},{2스포츠-시즌스포츠-남성수영복}> 0.0017012188                2
18                                             <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-남성비치웨어}> 0.0016778977                2
19                                           <{1의류-유아동의류-남아티셔츠/탑},{2스포츠-시즌스포츠-아동수영복}> 0.0016116164                2
20                                             <{2스포츠-시즌스포츠-아동수영복},{1의류-여성의류-여성티셔츠/탑}> 0.0015355157                2
21                                             <{1의류-여성의류-여성남방셔츠},{2스포츠-시즌스포츠-아동수영복}> 0.0015269237                2
22                                           <{2스포츠-시즌스포츠-남성수영복},{2스포츠-시즌스포츠-남성수영복}> 0.0015171042                2
23                                           <{3기타패션-패션잡화-유아동샌들},{2스포츠-시즌스포츠-아동수영복}> 0.0015121945                2
24                                           <{2스포츠-시즌스포츠-여성수영복},{2스포츠-시즌스포츠-여성수영복}> 0.0015060574                2
25                                         <{1의류-속옷/양말/홈웨어-유아동내의},{2스포츠-시즌스포츠-아동수영복}> 0.0014704619                2
26                                                 <{1의류-여성의류-여성바지},{2스포츠-시즌스포츠-아동수영복}> 0.0014459133                2
27                                           <{1의류-남성의류-남성캐주얼바지},{2스포츠-시즌스포츠-남성수영복}> 0.0014410035                2
28                                               <{2스포츠-시즌스포츠-아루아슈즈},{1의류-남성의류-남성티셔츠}> 0.0014397761                2
29                                         <{2스포츠-시즌스포츠-여성비치웨어},{2스포츠-시즌스포츠-여성비치웨어}> 0.0014311841                2
30                                             <{2스포츠-아웃도어/레저-등산화},{2스포츠-아웃도어/레저-등산화}> 0.0014029532                2
31                                            <{1의류-여성의류-여성티셔츠/탑},{2스포츠-시즌스포츠-아루아슈즈}> 0.0014004983                2
32                                           <{1의류-남성의류-남성캐주얼바지},{2스포츠-시즌스포츠-아동수영복}> 0.0013943612                2
33                                           <{1의류-여성의류-여성남방셔츠},{2스포츠-시즌스포츠-여성비치웨어}> 0.0013919063                2
34                                             <{1의류-유아동의류-영유아바지},{2스포츠-시즌스포츠-아동수영복}> 0.0013759497                2
35                                               <{2스포츠-시즌스포츠-남성수영복},{1의류-남성의류-남성티셔츠}> 0.0013538560                2
36                                           <{2스포츠-시즌스포츠-아동수영복},{1의류-유아동의류-영유아티셔츠/탑}> 0.0013440365                2
37                                             <{1의류-여성의류-여성남방셔츠},{2스포츠-시즌스포츠-남성수영복}> 0.0013342171                2
38                                               <{1의류-여성의류-여성원피스},{2스포츠-시즌스포츠-아루아슈즈}> 0.0013317622                2
39                                             <{2스포츠-시즌스포츠-아루아슈즈},{2스포츠-시즌스포츠-아루아슈즈}> 0.0013047587                2
40                                       <{2스포츠-구기/필드스포츠-골프필드용품},{2스포츠-구기/필드스포츠-골프필드용품}> 0.0012924844                2
41                                             <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-기타물놀이용품}> 0.0012568890                2
42                                           <{2스포츠-시즌스포츠-아동수영복},{1의류-유아동의류-여아티셔츠/탑}> 0.0012568890                2
43                                               <{1의류-여성의류-여성바지},{2스포츠-시즌스포츠-여성비치웨어}> 0.0012544341                2
44                                                 <{1의류-여성의류-여성바지},{2스포츠-시즌스포츠-남성수영복}> 0.0012421598                2
45                                             <{2스포츠-시즌스포츠-여성비치웨어},{1의류-여성의류-여성원피스}> 0.0012409324                2
46                                               <{1의류-남성의류-남성티셔츠},{2스포츠-아웃도어/레저-등산화}> 0.0012163837                2
47                                               <{2스포츠-시즌스포츠-아루아슈즈},{1의류-여성의류-여성원피스}> 0.0012102466                2
48                                             <{2스포츠-시즌스포츠-아동수영복},{1의류-남성의류-남성캐주얼바지}> 0.0012041094                2
49                                       <{2스포츠-시즌스포츠-아동수영복},{1의류-스포츠패션-유아동스포츠샌들/슬리퍼}> 0.0011844705                2
50                                               <{1의류-남성의류-남성티셔츠},{2스포츠-시즌스포츠-여성수영복}> 0.0011758785                2
51                                               <{2스포츠-아웃도어/레저-등산화},{1의류-남성의류-남성티셔츠}> 0.0011709688                2
52                                             <{1의류-남성의류-남성티셔츠},{2스포츠-구기/필드스포츠-골프필드용품}> 0.0011660591                2
53                                             <{2스포츠-구기/필드스포츠-골프필드용품},{1의류-남성의류-남성티셔츠}> 0.0011415105                2
```

사실 결과가 더 많았고, 그 결과중 고른 규칙들도 있지만, 어떤형식으로 결과가 나오는지만 보여주기 위하여 부분만 캡처함

3패션잡화

seqrule_fashion<-seqrule_subset2[grep("3",seqrule_subset2$sequence),]

suqrule_fashion_desc<-seqrule_fashion%>%arrange(desc(support))

suqrule_fashion_desc[1:300,]

	sequence	support	pms_seq_rule_size
1	<{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성샌들}>	0.0042469099	2
2	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성원피스}>	0.0041069829	2
3	<{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-여성샌들}>	0.0038553596	2
4	<{1의류-여성의류-여성남방셔츠},{3기타패션-패션잡화-여성샌들}>	0.0037669846	2
5	<{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-여성샌들}>	0.0037620748	2
6	<{3기타패션-패션잡화-여성샌들},{3기타패션-패션잡화-여성샌들}>	0.0037620748	2
7	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성남방셔츠}>	0.0030366634	2
8	<{1의류-여성의류-여성바지},{3기타패션-패션잡화-여성샌들}>	0.0030243890	2
9	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성티셔츠/탑}>	0.0029470609	2
10	<{3기타패션-패션잡화-여성샌들},{1의류-남성의류-남성티셔츠}>	0.0028918265	2
11	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성바지}>	0.0026696616	2
12	<{3기타패션-패션잡화-유아동샌들},{3기타패션-패션잡화-유아동샌들}>	0.0023713960	2
13	<{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-유아동샌들}>	0.0021565956	2
14	<{1의류-여성의류-여성스커트},{3기타패션-패션잡화-여성샌들}>	0.0020338525	2
15	<{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-여성슬더백}>	0.0020056216	2
16	<{3기타패션-패션잡화-여성펠트스},{1의류-여성의류-여성원피스}>	0.0019945748	2
17	<{1의류-남성의류-남성캐주얼바지},{3기타패션-패션잡화-여성샌들}>	0.0019663439	2
18	<{3기타패션-패션잡화-유아동샌들},{1의류-남성의류-남성티셔츠}>	0.0019049723	2
19	<{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-여성슬더백}>	0.0018350088	2
20	<{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성샌들}>	0.0018190522	3
21	<{3기타패션-패션잡화-여성슬더백},{1의류-남성의류-남성티셔츠}>	0.0017945036	2
22	<{3기타패션-패션잡화-여성샌들},{1의류-남성의류-남성캐주얼바지}>	0.0017883664	2
23	<{3기타패션-패션잡화-여성스니커즈},{1의류-남성의류-남성티셔츠}>	0.0017846841	2
24	<{1의류-유아동의류-여아티셔츠/탑},{3기타패션-패션잡화-유아동샌들}>	0.0017797744	2
25	<{3기타패션-패션잡화-여성슬더백},{1의류-여성의류-여성원피스}>	0.0017601355	2
26	<{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-여성스니커즈}>	0.0017380418	2
27	<{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성슬더백}>	0.0017368143	2
28	<{3기타패션-패션잡화-여성펠트스},{1의류-여성의류-여성남방셔츠}>	0.0017269949	2
29	<{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성원피스}>	0.0016852622	3
30	<{3기타패션-패션잡화-유아동샌들},{1의류-여성의류-여성원피스}>	0.0016668508	2
31	<{1의류-여성의류-여성남방셔츠},{3기타패션-패션잡화-여성슬더백}>	0.0016619411	2
32	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성원피스},{1의류-여성의류-여성원피스}>	0.0016558039	3
33	<{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-유아동샌들}>	0.0016324827	2
34	<{3기타패션-패션잡화-여성슬더백},{1의류-여성의류-여성티셔츠/탑}>	0.0016128438	2
35	<{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-여성슬리퍼}>	0.0015968872	2
36	<{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성슬리퍼}>	0.0015907501	2
37	<{3기타패션-패션잡화-여성샌들},{1의류-여성의류-여성스커트}>	0.0015858404	2
38	<{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-여성슬리퍼}>	0.0015723386	2
39	<{3기타패션-패션잡화-여성스니커즈},{1의류-여성의류-여성원피스}>	0.0015637466	2
40	<{4뷰티건강-화장품/뷰티케어-BB/파운데이션/컴팩트류},{3기타패션-패션잡화-여성샌들}>	0.0015625192	2
41	<{1의류-유아동의류-영유아티셔츠/탑},{3기타패션-패션잡화-유아동샌들}>	0.0015490174	2
42	<{3기타패션-패션잡화-여성슬더백},{3기타패션-패션잡화-여성슬더백}>	0.0015121945	2
43	<{3기타패션-패션잡화-유아동샌들},{2스포츠-시즌스포츠-아동수영복}>	0.0015121945	2
44	<{3기타패션-패션잡화-유아동샌들},{1의류-여성의류-여성티셔츠/탑}>	0.0015023751	2
45	<{1의류-여성의류-여성남방셔츠},{1의류-여성의류-여성남방셔츠},{3기타패션-패션잡화-여성샌들}>	0.0015011476	3
46	<{1의류-남성의류-남성티셔츠},{1의류-남성의류-남성티셔츠},{3기타패션-패션잡화-여성샌들}>	0.0014950105	3
47	<{3기타패션-패션잡화-여성스니커즈},{1의류-여성의류-여성티셔츠/탑}>	0.0014864185	2
48	<{1의류-여성의류-여성원피스},{3기타패션-패션잡화-여성펠트스}>	0.0014802813	2
49	<{3기타패션-패션잡화-여성슬더백},{1의류-여성의류-여성남방셔츠}>	0.0014532779	2
50	<{1의류-여성의류-여성재킷},{3기타패션-패션잡화-여성샌들}>	0.0014483681	2
51	<{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-여성스니커즈}>	0.0014238195	2
52	<{1의류-여성의류-여성티셔츠/탑},{1의류-여성의류-여성티셔츠/탑},{3기타패션-패션잡화-여성샌들}>	0.0014238195	3
53	<{3기타패션-패션잡화-여성슬리퍼},{1의류-여성의류-여성원피스}>	0.0014225921	2

사실 결과가 더 많았고, 그 결과중 고른 규칙들도 있지만, 어떤형식으로 결과가 나오는지만 보여주기 위하여 부분만 캡처함

4BH

```
seqrule_bh<-seqrule_subset2[grep("4",seqrule_subset2$sequence),]
suqrule_bh_desc<-seqrule_bh%>%arrange(desc(support))
suqrule_bh_desc[1:200,]
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사실 결과가 더 많았고, 그 결과중 고른 규칙들도 있지만, 어떤형식으로 결과가 나오는지만 보여주기 위하여 부분만 캡처함

5BABY

seqrule_baby<-seqrule_subset2[grep("5",seqrule_subset2$sequence),]

suqrule_baby_desc<-seqrule_baby%>%arrange(desc(support))

suqrule_baby_desc[1:100,]

```
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
sequence      support pms_seq_rule_size
1                    <{5출산육아-출산/육아용품-유아용화장품},{5출산육아-출산/육아용품-유아용화장품}> 0.0023394828                 2
2                          <{1의류-남성의류-남성티셔츠},{5출산육아-출산/육아용품-유아용화장품}> 0.0019381129                 2
3                    <{5출산육아-출산/육아용품-유아용물티슈},{5출산육아-출산/육아용품-유아용물티슈}> 0.0017834567                 2
4                          <{5출산육아-출산/육아용품-유아용화장품},{1의류-남성의류-남성티셔츠}> 0.0017478612                 2
5                          <{5출산육아-출산/육아용품-유아용화장품},{1의류-여성의류-여성원피스}> 0.0016054793                 2
6                        <{1의류-여성의류-여성티셔츠/탑},{5출산육아-출산/육아용품-유아용화장품}> 0.0015330608                 2
7                        <{5출산육아-출산/육아용품-유아용화장품},{1의류-여성의류-여성티셔츠/탑}> 0.0014729167                 2
8                      <{5출산육아-출산/육아용품-이유식용품},{5출산육아-출산/육아용품-이유식용품}> 0.0014250469                 2
9                          <{1의류-여성의류-여성원피스},{5출산육아-출산/육아용품-유아용화장품}> 0.0014078629                 2
10                   <{5출산육아-출산/육아용품-유아용기저귀},{5출산육아-출산/육아용품-유아용기저귀}> 0.0013796320                 2
11                         <{1의류-남성의류-남성티셔츠},{5출산육아-출산/육아용품-유아용물티슈}> 0.0012335678                 2
12                     <{1의류-속옷/양말/홈웨어-유아동내의},{5출산육아-출산/육아용품-유아용화장품}> 0.0012114740                 2
13                        <{1의류-유아동의류-여아티셔츠/탑},{5출산육아-출산/육아용품-유아용화장품}> 0.0011967449                 2
14                               <{5출산육아-출산/육아용품-유아용화장품},{1의류-여성의류-여성바지}> 0.0011648317                 2
15                      <{1의류-유아동의류-영유아티셔츠/탑},{5출산육아-출산/육아용품-유아용화장품}> 0.0011194167                 2
16                            <{5출산육아-출산/육아용품-유아용화장품},{1의류-여성의류-여성남방셔츠}> 0.0010899584                 2
17                             <{5출산육아-출산/육아용품-유아용물티슈},{1의류-남성의류-남성티셔츠}> 0.0010605001                 2
18                               <{1의류-여성의류-여성바지},{5출산육아-출산/육아용품-유아용화장품}> 0.0010445435                 2
19                     <{5출산육아-출산/육아용품-유아용화장품},{1의류-속옷/양말/홈웨어-유아동내의}> 0.0010420886                 2
20                        <{1의류-여성의류-여성티셔츠/탑},{5출산육아-출산/육아용품-유아용물티슈}> 0.0010408612                 2
21                          <{1의류-여성의류-여성남방셔츠},{5출산육아-출산/육아용품-유아용화장품}> 0.0010371789                 2
22                        <{5출산육아-출산/육아용품-유아용화장품},{2스포츠-시즌스포츠-아동수영복}> 0.0010224497                 2
23                        <{5출산육아-출산/육아용품-유아용화장품},{1의류-남성의류-남성캐주얼바지}> 0.0010150851                 2
24                  <{5출산육아-출산/육아용품-유아용화장품},{4뷰티건강-화장품/뷰티케어-스킨케어세트}> 0.0010040382                 2
25                  <{4뷰티건강-화장품/뷰티케어-스킨케어세트},{5출산육아-출산/육아용품-유아용화장품}> 0.0009954462                 2
26                       <{5출산육아-출산/육아용품-유아용화장품},{1의류-유아동의류-여아티셔츠/탑}> 0.0009770348                 2
27                <{5출산육아-출산/육아용품-유아용화장품},{1의류-스포츠패션-유아동스포츠샌들/슬리퍼}> 0.0009721251                 2
28                     <{5출산육아-출산/육아용품-유아용화장품},{1의류-유아동의류-영유아티셔츠/탑}> 0.0009488039                 2
29                          <{1의류-여성의류-여성원피스},{5출산육아-출산/육아용품-유아용물티슈}> 0.0009402119                 2
30                <{1의류-스포츠패션-유아동스포츠샌들/슬리퍼},{5출산육아-출산/육아용품-유아용화장품}> 0.0009365296                 2
31                          <{5출산육아-출산/육아용품-유아용물티슈},{1의류-여성의류-여성원피스}> 0.0009303924                 2
32                          <{1의류-유아동의류-남아바지},{5출산육아-출산/육아용품-유아용화장품}> 0.0009291650                 2
33        <{5출산육아-출산/육아용품-유아용화장품},{4뷰티건강-화장품/뷰티케어-BB/파운데이션/컴팩트류}> 0.0009254827                 2
34                        <{1의류-남성의류-남성캐주얼바지},{5출산육아-출산/육아용품-유아용화장품}> 0.0009119810                 2
35      <{5출산육아-출산/육아용품-출산/신생아용품세트},{5출산육아-출산/육아용품-출산/신생아용품세트}> 0.0009033889                 2
36                        <{5출산육아-출산/육아용품-유아용화장품},{3기타패션-패션잡화-유아동샌들}> 0.0008923421                 2
37                      <{1의류-속옷/양말/홈웨어-유아동내의},{5출산육아-출산/육아용품-이유식용품}> 0.0008788403                 2
38                         <{5출산육아-출산/육아용품-유아용물티슈},{1의류-여성의류-여성티셔츠/탑}> 0.0008776129                 2
39                        <{3기타패션-패션잡화-유아동샌들},{5출산육아-출산/육아용품-유아용화장품}> 0.0008677934                 2
40                               <{1의류-남성의류-남성티셔츠},{5출산육아-출산/육아용품-이유식용품}> 0.0008530643                 2
41        <{4뷰티건강-화장품/뷰티케어-BB/파운데이션/컴팩트류},{5출산육아-출산/육아용품-유아용화장품}> 0.0008371077                 2
42                  <{5출산육아-출산/육아용품-유아용화장품},{5출산육아-출산/육아용품-유아용물티슈}> 0.0007990573                 2
43                  <{5출산육아-출산/육아용품-유아용화장품},{4뷰티건강-화장품/뷰티케어-페이셜클렌저}> 0.0007831007                 2
44                   <{1의류-속옷/양말/홈웨어-유아동일반양말},{5출산육아-출산/육아용품-유아용화장품}> 0.0007818733                 2
45                          <{5출산육아-출산/육아용품-이유식용품},{1의류-남성의류-남성티셔츠}> 0.0007745087                 2
46                     <{5출산육아-출산/육아용품-이유식용품},{5출산육아-출산/육아용품-유아용화장품}> 0.0007695990                 2
47                     <{5출산육아-출산/육아용품-젖병/젖꼭지},{5출산육아-출산/육아용품-젖병/젖꼭지}> 0.0007671441                 2
48                       <{1의류-남성의류-남성티셔츠},{5출산육아-출산/육아용품-출산/신생아용품세트}> 0.0007634618                 2
49                               <{1의류-여성의류-여성바지},{5출산육아-출산/육아용품-유아용물티슈}> 0.0007634618                 2
50                       <{5출산육아-출산/육아용품-유아용화장품},{1의류-속옷/양말/홈웨어-유아동팬티}> 0.0007597796                 2
51                          <{2스포츠-시즌스포츠-아동수영복},{5출산육아-출산/육아용품-유아용화장품}> 0.0007573247                 2
52                        <{1의류-유아동의류-남아티셔츠/탑},{5출산육아-출산/육아용품-유아용화장품}> 0.0007560973                 2
53                          <{5출산육아-출산/육아용품-유아용화장품},{1의류-유아동의류-남아바지}> 0.0007560973                 2
```

사실 결과가 더 많았고, 그 결과중 고른 규칙들도 있지만, 어떤형식으로 결과가 나오는지만 보여주기 위하여 부분만 캡처함

6HOME

```
seqrule_home<-seqrule_subset2[grep("6",seqrule_subset2$sequence),]
suqrule_home_desc<-seqrule_home%>%arrange(desc(support))
suqrule_home_desc[1:100,]
```

	sequence	support	pms_seq_rule_size
1	<{1의류-남성의류-남성티셔츠},{6홈케어-세제/위생-액상세탁세제}>	0.0013599931	2
2	<{1의류-남성의류-남성티셔츠},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0013526285	2
3	<{1의류-여성의류-여성티셔츠/탑},{6홈케어-세제/위생-액상세탁세제}>	0.0011255539	2
4	<{1의류-여성의류-여성티셔츠/탑},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0011218716	2
5	<{6홈케어-세제/위생-액상세탁세제},{1의류-남성의류-남성티셔츠}>	0.0010948681	2
6	<{6홈케어-세제/위생-섬유유연제/향기지속제},{1의류-남성의류-남성티셔츠}>	0.0010801389	2
7	<{6홈케어-세제/위생-액상세탁세제},{6홈케어-세제/위생-액상세탁세제}>	0.0010519080	2
8	<{6홈케어-세제/위생-섬유유연제/향기지속제},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0010298143	2
9	<{1의류-남성의류-남성티셔츠},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0010212223	2
10	<{1의류-남성의류-남성티셔츠},{6홈케어-식기/조리기구-프라이팬}>	0.0010064931	2
11	<{1의류-여성의류-여성원피스},{6홈케어-세제/위생-액상세탁세제}>	0.0009721251	2
12	<{1의류-남성의류-남성티셔츠},{6홈케어-주방잡화-기타주방정리용품/소모품}>	0.0009672153	2
13	<{1의류-여성의류-여성원피스},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0009537136	2
14	<{6홈케어-세제/위생-액상세탁세제},{1의류-여성의류-여성원피스}>	0.0008923421	2
15	<{6홈케어-세제/위생-섬유유연제/향기지속제},{1의류-여성의류-여성원피스}>	0.0008874323	2
16	<{6홈케어-세제/위생-섬유유연제/향기지속제},{1의류-여성의류-여성티셔츠/탑}>	0.0008837500	2
17	<{6홈케어-식기/조리기구-프라이팬},{1의류-남성의류-남성티셔츠}>	0.0008800678	2
18	<{6홈케어-세제/위생-액상세탁세제},{1의류-여성의류-여성티셔츠/탑}>	0.0008714757	2
19	<{1의류-여성의류-여성티셔츠/탑},{6홈케어-주방잡화-기타주방정리용품/소모품}>	0.0008665660	2
20	<{1의류-여성의류-여성티셔츠/탑},{6홈케어-식기/조리기구-프라이팬}>	0.0008260608	2
21	<{1의류-여성의류-여성티셔츠/탑},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0008223785	2
22	<{1의류-여성의류-여성남방셔츠},{6홈케어-세제/위생-액상세탁세제}>	0.0008211511	2
23	<{6홈케어-식기/조리기구-반찬통/밀폐용기},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0008186962	2
24	<{1의류-여성의류-여성남방셔츠},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0008174688	2
25	<{6홈케어-주방잡화-기타주방정리용품/소모품},{1의류-남성의류-남성티셔츠}>	0.0008125591	2
26	<{1의류-여성의류-여성바지},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0008076493	2
27	<{6홈케어-식기/조리기구-반찬통/밀폐용기},{1의류-남성의류-남성티셔츠}>	0.0008015122	2
28	<{1의류-여성의류-여성바지},{6홈케어-세제/위생-액상세탁세제}>	0.0007757362	2
29	<{6홈케어-식기/조리기구-접시},{6홈케어-식기/조리기구-접시}>	0.0007646893	2
30	<{1의류-여성의류-여성원피스},{6홈케어-식기/조리기구-프라이팬}>	0.0007622344	2
31	<{6홈케어-세제/위생-액상세탁세제},{1의류-여성의류-여성남방셔츠}>	0.0007352309	2
32	<{6홈케어-식기/조리기구-반찬통/밀폐용기},{1의류-여성의류-여성원피스}>	0.0007254115	2
33	<{1의류-여성의류-여성원피스},{6홈케어-주방잡화-기타주방정리용품/소모품}>	0.0007241841	2
34	<{1의류-여성의류-여성원피스},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0007241841	2
35	<{1의류-남성의류-남성캐주얼바지},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0007192743	2
36	<{6홈케어-세제/위생-섬유유연제/향기지속제},{1의류-여성의류-여성남방셔츠}>	0.0007119098	2
37	<{6홈케어-주방잡화-기타주방정리용품/소모품},{1의류-여성의류-여성원피스}>	0.0007082275	2
38	<{6홈케어-식기/조리기구-프라이팬},{1의류-여성의류-여성원피스}>	0.0007033177	2
39	<{6홈케어-세제/위생-섬유유연제/향기지속제},{1의류-여성의류-여성바지}>	0.0007020903	2
40	<{1의류-남성의류-남성캐주얼바지},{6홈케어-세제/위생-액상세탁세제}>	0.0006984080	2
41	<{6홈케어-세제/위생-액상세탁세제},{6홈케어-세제/위생-섬유유연제/향기지속제}>	0.0006959532	2
42	<{6홈케어-세제/위생-액상세탁세제},{1의류-여성의류-여성바지}>	0.0006934983	2
43	<{6홈케어-주방잡화-기타주방정리용품/소모품},{1의류-여성의류-여성티셔츠/탑}>	0.0006873611	2
44	<{1의류-여성의류-여성남방셔츠},{6홈케어-식기/조리기구-프라이팬}>	0.0006861337	2
45	<{1의류-여성의류-여성바지},{6홈케어-식기/조리기구-프라이팬}>	0.0006861337	2
46	<{6홈케어-식기/조리기구-프라이팬},{1의류-여성의류-여성티셔츠/탑}>	0.0006750868	2
47	<{6홈케어-식기/조리기구-반찬통/밀폐용기},{1의류-여성의류-여성티셔츠/탑}>	0.0006738594	2
48	<{6홈케어-세제/위생-액상세탁세제},{1의류-남성의류-남성캐주얼바지}>	0.0006615851	2
49	<{6홈케어-세제/위생-섬유유연제/향기지속제},{6홈케어-세제/위생-액상세탁세제}>	0.0006529931	2
50	<{1의류-남성의류-남성티셔츠},{6홈케어-세제/위생-롤티슈}>	0.0006517657	2
51	<{1의류-여성의류-여성바지},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0006468559	2
52	<{1의류-여성의류-여성남방셔츠},{6홈케어-식기/조리기구-반찬통/밀폐용기}>	0.0006444011	2
53	<{1의류-남성의류-남성티셔츠},{6홈케어-식기/조리기구-접시}>	0.0006419462	2
54	<{1의류-여성의류-여성남방셔츠},{6홈케어-주방잡화-기타주방정리용품/소모품}>	0.0006370365	2

Ⅲ-(2) 검색어를 통한 트렌드 예측

Search1으로 고객 구매예측

원피스

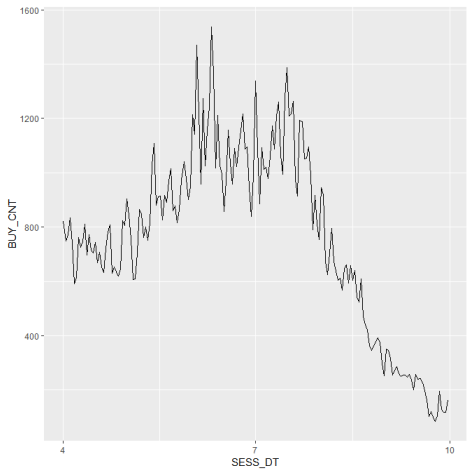
원피스라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

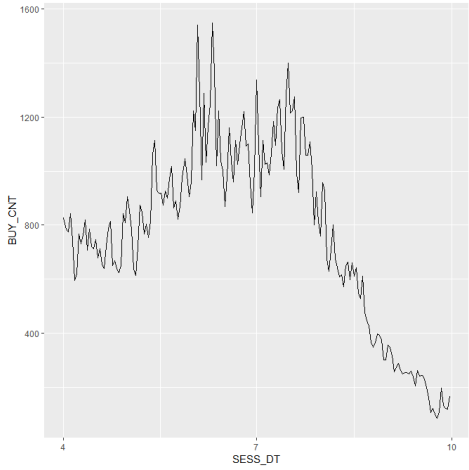
```
search1<-read_csv("search1.csv")
custom<-read_rds("custom.rds")
search1$CLNT_ID<-as.character(search1$CLNT_ID)
s1c<-inner_join(search1,custom)
s1c_F30<-s1c%>%filter(CLNT_GENDER=="F",CLNT_AGE==30)
s1c_F30%>%count(KWD_NM,SEARCH_CNT)%>%arrange(desc(n))
```

```
# A tibble: 158,759 x 3
   KWD_NM          SEARCH_CNT     n
   <chr>                <dbl> <int>
 1 뉴발란스키즈             1  1718
 2 원피스                   1  1301
 3 아디다스키즈             1  1285
 4 블루독                   1  1261
 5 생수                     1  1084
 6 노스페이스키즈           1  1042
 7 크록스                   1   945
 8 블루독베이비             1   877
 9 임생로랑                 1   842
10 여성샌들                 1   826
# ... with 158,749 more rows
```

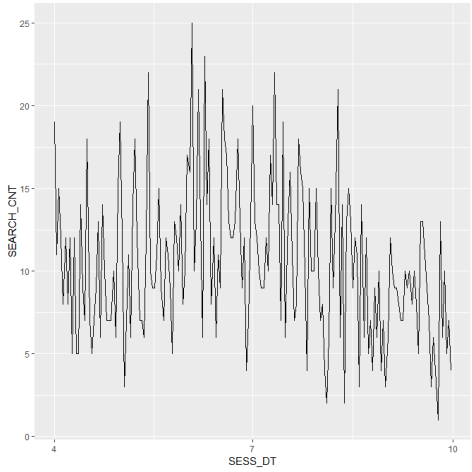
```
master<-read_rds("master.rds")
master_원피스<-master%>%filter(str_detect(CLAC3_NM,"원피스"))
product<-read_rds("product.rds")
session<-read_rds("session.rds")
ps<-inner_join(product,session)
psm_원피스_pre<-inner_join(ps,master_원피스)
psm_원피스<-psm_원피스
_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())%>%ungroup()
psm_원피스_trans<-psm_원피스
_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))%>%ungroup()
s1cs_F30<-inner_join(s1c_F30,session)
s1_F30원피스<-s1cs_F30%>%filter(KWD_NM=="원피스
",CLNT_GENDER=="F",CLNT_AGE==30)%>%group_by(SESS_DT)%>%summarize(SEARCH_CNT=sum
(SEARCH_CNT))%>%ungroup()
psm_원피스%>%ggplot(aes(SESS_DT,BUY_CNT))+geom_line()
```



psm_원피스_trans%>%ggplot(aes(SESS_DT,BUY_CNT))+geom_line()



s1_F30원피스%>%ggplot(aes(SESS_DT,SEARCH_CNT))+geom_line()

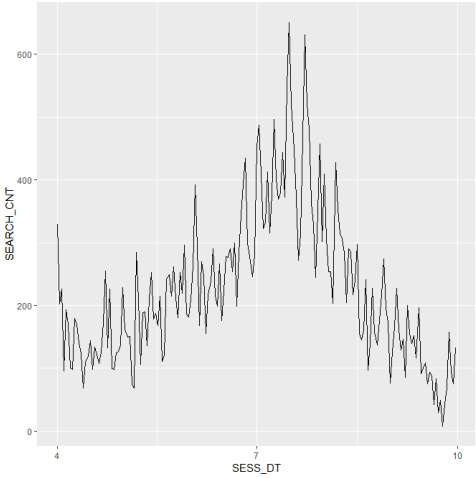


수영

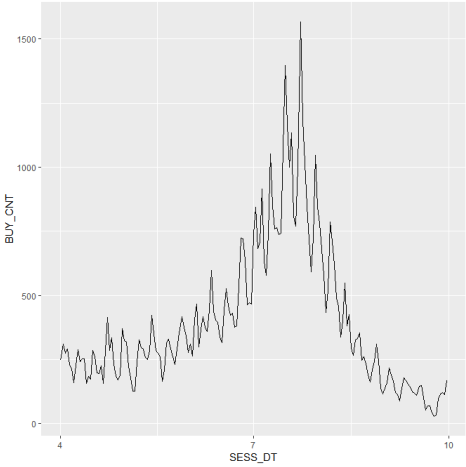
수영이라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

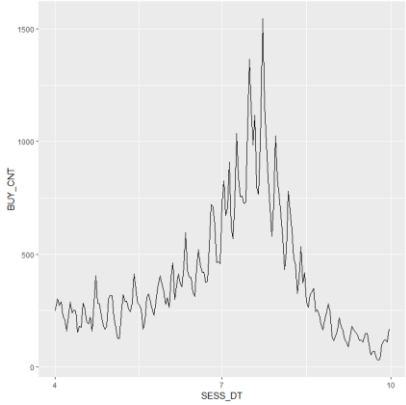
```
s1_수영<-search1%>%filter(str_detect(KWD_NM,"수영"))
s1s_수영<-inner_join(s1_수영,session)
s1s_수영_cnt<-s1s_수영%>%group_by(SESS_DT)%>%summarize(SEARCH_CNT=sum(SEARCH_CNT))
pm<-inner_join(product,master)
pm_수영_pre<-pm%>%filter(str_detect(CLAC3_NM,"수영"))
pms_수영_pre<-inner_join(pm_수영_pre,session)
pm_수영<-pms_수영_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))
pm_수영_trans<-pms_수영_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())
ggplot(s1s_수영_cnt,aes(SESS_DT,SEARCH_CNT))+geom_line()
```




```
ggplot(pm_수영,aes(SESS_DT,BUY_CNT))+geom_line()
```




```
ggplot(pm_수영_trans,aes(SESS_DT,BUY_CNT))+geom_line()
```

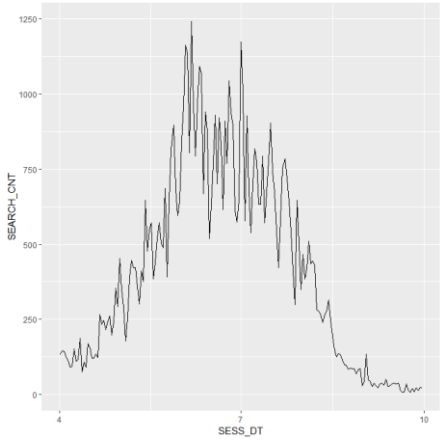


샌들

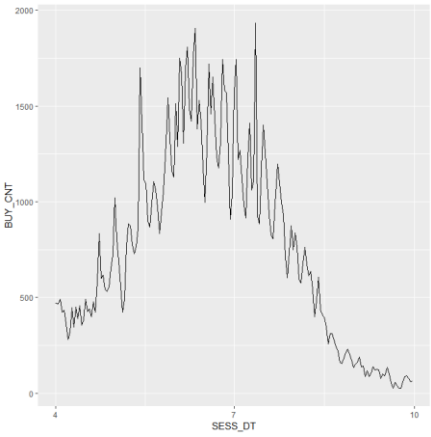
샌들이라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

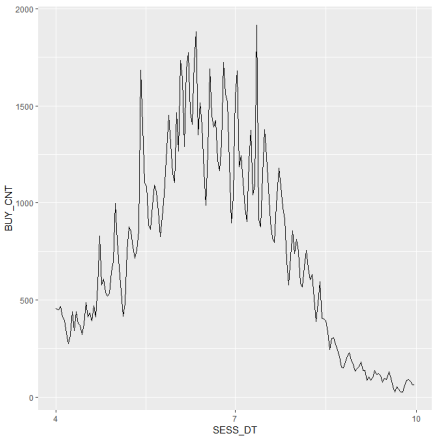
```
s1_샌들<-search1%>%filter(str_detect(KWD_NM,"샌들"))
s1s_샌들<-inner_join(s1_샌들,session)
s1s_샌들_cnt<-s1s_샌들%>%group_by(SESS_DT)%>%summarize(SEARCH_CNT=sum(SEARCH_CNT))
pm<-inner_join(product,master)
pm_샌들_pre<-pm%>%filter(str_detect(CLAC3_NM,"샌들"))
pms_샌들_pre<-inner_join(pm_샌들_pre,session)
pm_샌들<-pms_샌들_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))
pm_샌들_trans<-pms_샌들_pre%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())
ggplot(s1s_샌들_cnt,aes(SESS_DT,SEARCH_CNT))+geom_line()
```




```
ggplot(pm_샌들,aes(SESS_DT,BUY_CNT))+geom_line()
```



ggplot(pm_샌들_trans,aes(SESS_DT,BUY_CNT))+geom_line()



Search2로 고객 구매예측

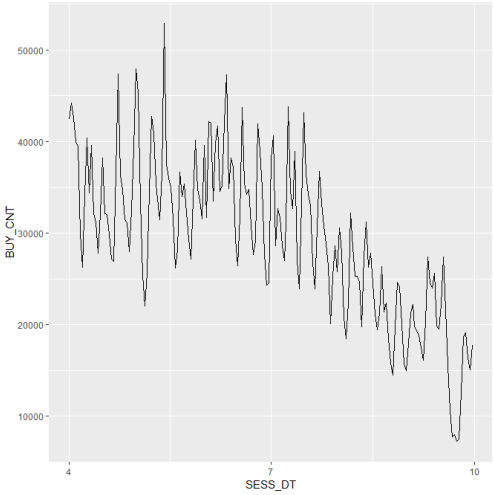
원피스

원피스이라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

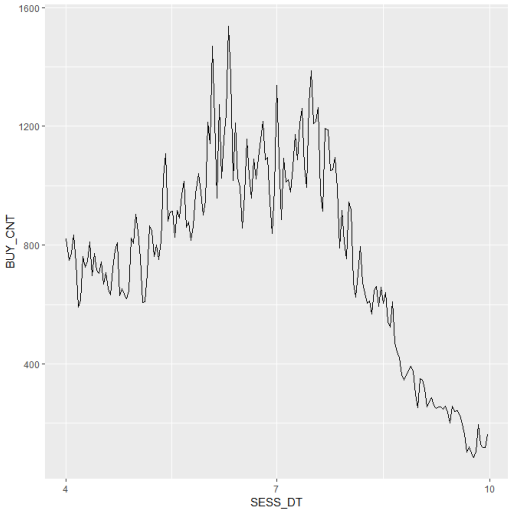
str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

```
search2<-read_rds("search2.rds")
product<-read_rds("product.rds")
master<-read_rds("master.rds")
session<-read_rds("session.rds")
s2_원피스<-search2%>%filter(str_detect(KWD_NM,"원피스"))
s2_원피스<-s2_원피
스%>%group_by(SESS_DT,SESS_DAY)%>%summarize(SEARCH_CNT=sum(SEARCH_CNT))%>%ungro
up()
pm<-inner_join(product,master)
pms<-inner_join(pm,session)
pms_원피스<-pms%>%filter(CLAC3_NM=="여성원피스
")%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))
```

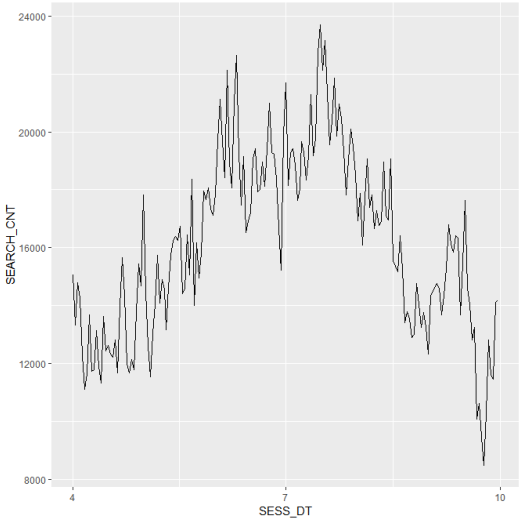
```
pms_원피스_trans<-pms%>%filter(CLAC3_NM=="여성원피스
")%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())
ggplot(pms_원피스,aes(SESS_DT,BUY_CNT))+geom_line()
```




```
ggplot(pms_원피스_trans,aes(SESS_DT,BUY_CNT))+geom_line()
```




```
ggplot(s2_원피스,aes(SESS_DT,SEARCH_CNT))+geom_line()
```



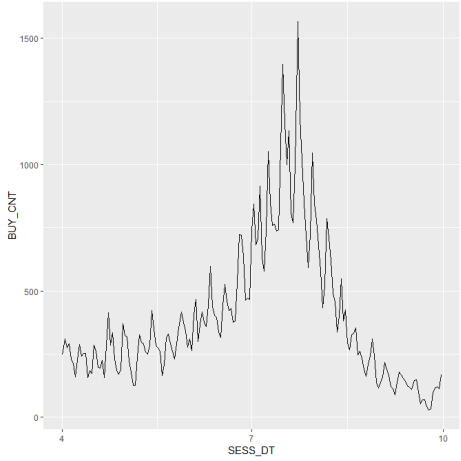
계절용인 수영도 체크해보자!

수영

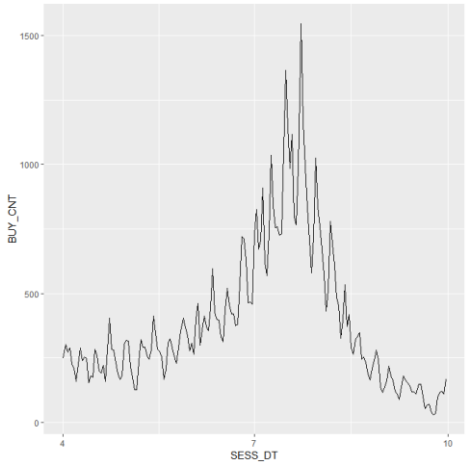
수영이라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

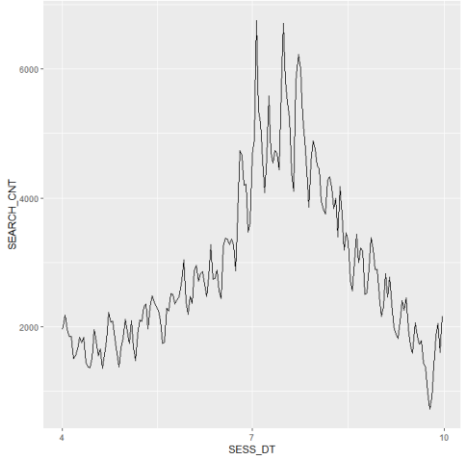
```
search2<-read_rds("search2.rds")
product<-read_rds("product.rds")
master<-read_rds("master.rds")
session<-read_rds("session.rds")
s2_수영<-search2%>%filter(str_detect(KWD_NM,"수영"))
s2_수영<-s2_수
영%>%group_by(SESS_DT,SESS_DAY)%>%summarize(SEARCH_CNT=sum(SEARCH_CNT))%>%ungro
up()
pm<-inner_join(product,master)
pms<-inner_join(pm,session)
pms_수영<-pms%>%filter(str_detect(CLAC3_NM,"수영
"))%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))
pms_수영_trans<-pms%>%filter(str_detect(CLAC3_NM,"수영
"))%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())
ggplot(pms_수영,aes(SESS_DT,BUY_CNT))+geom_line()
```



ggplot(pms_수영_trans,aes(SESS_DT,BUY_CNT))+geom_line()



ggplot(s2_수영,aes(SESS_DT,SEARCH_CNT))+geom_line()



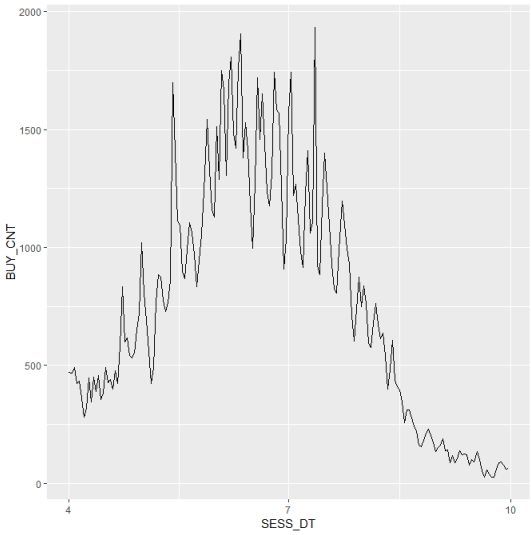
샌들

샌들이라는 단어가 들어간 검색어와 CLAC3_NM을 골라서 분석함.

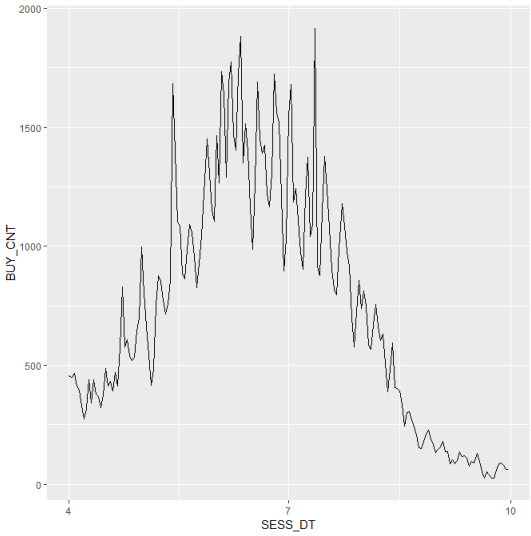
str_detect(변수,"z") 변수 obs 중에서 z가 들어간 것 logical로 답변.

search2<-read_rds("search2.rds")

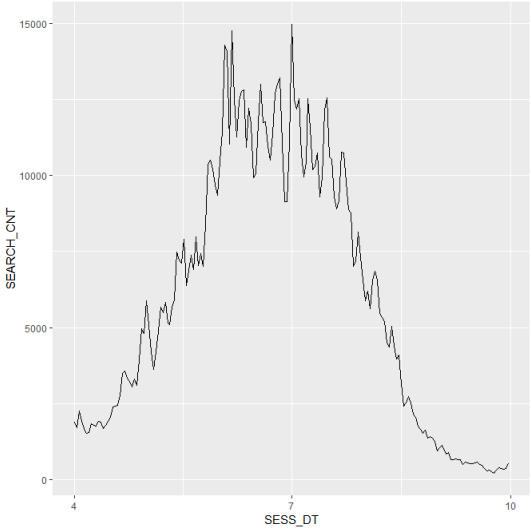
```
product<-read_rds("product.rds")
master<-read_rds("master.rds")
session<-read_rds("session.rds")
s2_샌들<-search2%>%filter(str_detect(KWD_NM,"샌들"))
s2_샌들<-s2_샌
들%>%group_by(SESS_DT,SESS_DAY)%>%summarize(SEARCH_CNT=sum(SEARCH_CNT))%>%ungro
up()
pm<-inner_join(product,master)
pms<-inner_join(pm,session)
pms_샌들<-pms%>%filter(str_detect(CLAC3_NM,"샌들
"))%>%group_by(SESS_DT)%>%summarize(BUY_CNT=sum(PD_BUY_CT))
pms_샌들_trans<-pms%>%filter(str_detect(CLAC3_NM,"샌들
"))%>%group_by(SESS_DT)%>%summarize(BUY_CNT=n())
ggplot(pms_샌들,aes(SESS_DT,BUY_CNT))+geom_line()
```




```
ggplot(pms_샌들_trans,aes(SESS_DT,BUY_CNT))+geom_line()
```



ggplot(s2_샌들,aes(SESS_DT,SEARCH_CNT))+geom_line()



IV. 서비스 제안

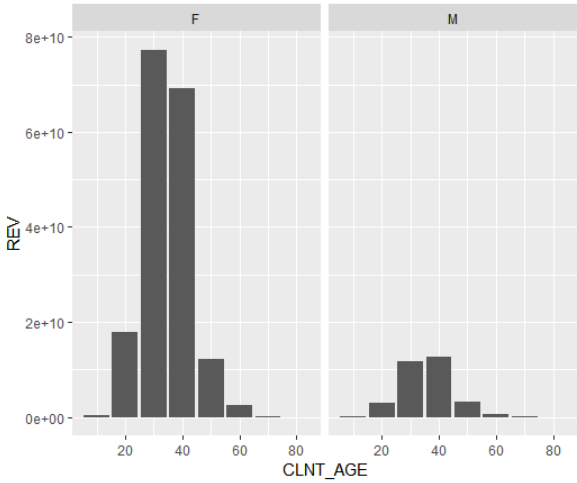
product<-read_rds("product.rds")

custom<-read_rds("custom.rds")

pc<-product%>%inner_join(custom)

pc_REV<-pc%>%group_by(CLNT_GENDER, CLNT_AGE)%>%summarize(REV=sum(PD_BUY_TOTAL))

ggplot(pc_REV,aes(CLNT_AGE, REV))+geom_bar(stat="identity")+facet_wrap(~CLNT_GENDER)



pc%>%group_by(CLNT_GENDER,

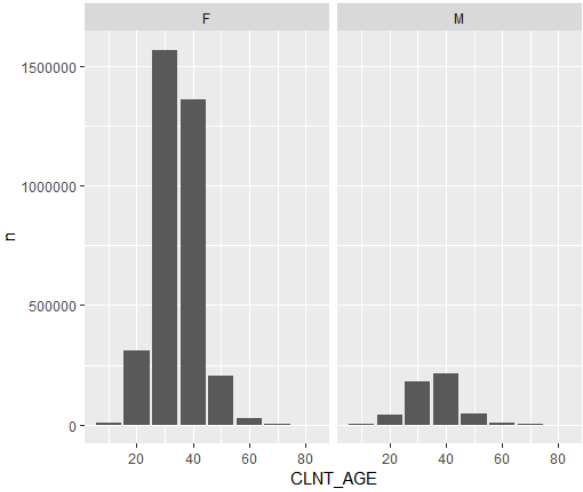
CLNT_AGE)%>%summarize(REV=sum(PD_BUY_TOTAL))%>%arrange(desc(REV))

```
# A tibble: 16 x 3
# Groups:   CLNT_GENDER [2]
   CLNT_GENDER CLNT_AGE         REV
   <chr>          <dbl>       <dbl>
 1 F                 30 77274160100
 2 F                 40 69289438960
 3 F                 20 17836310510
 4 M                 40 12609639740
 5 F                 50 12177132020
 6 M                 30 11703598871
 7 M                 50  3157926380
 8 M                 20  2966230490
 9 F                 60  2434191720
10 M                 60   503644390
11 F                 10   444122800
12 F                 70   190678580
13 M                 10   145106640
14 M                 70    81293510
15 F                 80     9955270
16 M                 80     3291400
```

매출액 높은 순서: F30 F40 F20 M40 F50 M30

pc_CNT<-pc%>%count(CLNT_GENDER,CLNT_AGE)%>%arrange(desc(n))

ggplot(pc_CNT, aes(CLNT_AGE,n))+geom_bar(stat="identity")+facet_wrap(~CLNT_GENDER)



pc_CNT

```
# A tibble: 16 x 3
   CLNT_GENDER CLNT_AGE       n
   <chr>          <dbl>   <int>
 1 F                 30 1567587
 2 F                 40 1360890
 3 F                 20  309989
 4 M                 40  212620
 5 F                 50  206491
 6 M                 30  181474
 7 M                 50   46057
 8 M                 20   43645
 9 F                 60   29086
10 F                 10    9662
11 M                 60    8179
12 F                 70    3700
13 M                 10    2440
14 M                 70    1762
15 F                 80     219
16 M                 80      36
```

구매횟수 많은 순서: F30 F40 F20 M40 F50 M30

master<-read_rds("master.rds")

pcm<-pc%>%left_join(master)

pcm1<-pcm%>%filter(CLAC1_NM %in% c("여성의류", "스포츠패션","남성의류","유아동의류","속옷/양말/홈웨어","시즌스포츠","구기/필드스포츠","아웃도어/레저","패션잡화","퍼스널케어","화장품/뷰티케어","건강식품","출산/육아용품","식기/조리기구","주방잡화","세제/위생"))

pcm1_1<-pcm1%>%filter(CLAC1_NM %in% c("여성의류","스포츠패션","남성의류","유아동의류","속옷/양말/홈웨어"))%>%mutate(PG="1의류")

pcm1_2<-pcm1%>%filter(CLAC1_NM %in% c("시즌스포츠","구기/필드스포츠","아웃도어/레저"))%>%mutate(PG="2스포츠")

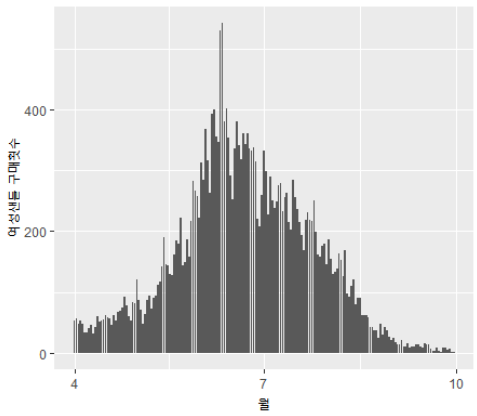
pcm1_3<-pcm1%>%filter(CLAC1_NM %in% c("패션잡화"))%>%mutate(PG="3기타패션")

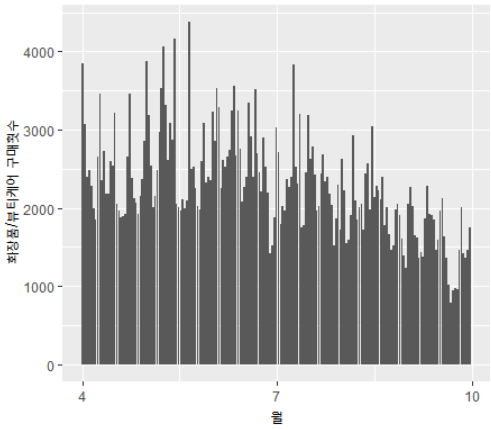
pcm1_4<-pcm1%>%filter(CLAC1_NM %in% c("퍼스널케어","화장품/뷰티케어","건강식품

```
"))%>%mutate(PG="4뷰티/건강")
pcm1_5<-pcm1%>%filter(CLAC1_NM %in% c("출산/육아용품"))%>%mutate(PG="5출산/육아")
pcm1_6<-pcm1%>%filter(CLAC1_NM %in% c("식기/조리기구","주방잡화","세제/위생
"))%>%mutate(PG="6홈케어")
pcm2<-union(pcm1_1,pcm1_2)
pcm2<-union(pcm2,pcm1_3)
pcm2<-union(pcm2,pcm1_4)
pcm2<-union(pcm2,pcm1_5)
pcm2<-union(pcm2,pcm1_6)
wirte_rds(pcm2,"pcm2.rds")
```

여성샌들과 화장품을 연속해서 사는 이유는?

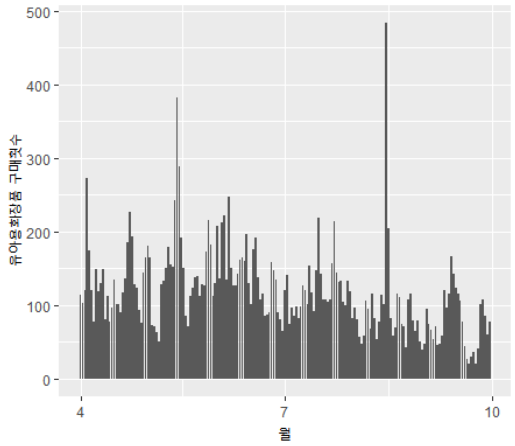
```
session<-read_rds("session.rds")
pcms<-inner_join(pcm,session)
pcms1<-pcms%>%filter(CLAC3_NM == "여성샌들"|CLAC1_NM=="화장품/뷰티케어")
pcms1_1<-pcms1%>%filter(CLAC3_NM=="여성샌들")%>%count(SESS_DT)

ggplot(pcms1_1, aes(SESS_DT, n))+geom_histogram(stat="identity")+xlab("월")+ylab("여성샌들 구매
횟수")
pcms1_2<-pcms1%>%filter(CLAC1_NM=="화장품/뷰티케어")%>%count(SESS_DT)
ggplot(pcms1_2, aes(SESS_DT, n))+geom_histogram(stat="identity")+xlab("월")+ylab("화장품/뷰티케
어 구매횟수")
```

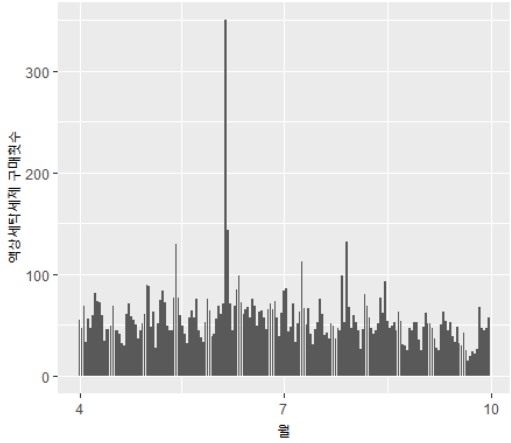




유아용 화장품과 액상세탁세제를 연속해서 사는 이유?

```
pcms2<-pcms%>%filter(CLAC3_NM %in% c("유아용화장품","액상세탁세제"))
pcms2_1<-pcms2%>%filter(CLAC3_NM=="유아용화장품")%>%count(SESS_DT)

ggplot(pcms2_1,aes(SESS_DT,n))+geom_histogram(stat="identity")+xlab("월")+ylab("유아용화장품 구매횟수")
pcms2_2<-pcms2%>%filter(CLAC3_NM=="액상세탁세제")%>%count(SESS_DT)
ggplot(pcms2_2,aes(SESS_DT,n))+geom_histogram(stat="identity")+xlab("월")+ylab("액상세탁세제 구매횟수")
```

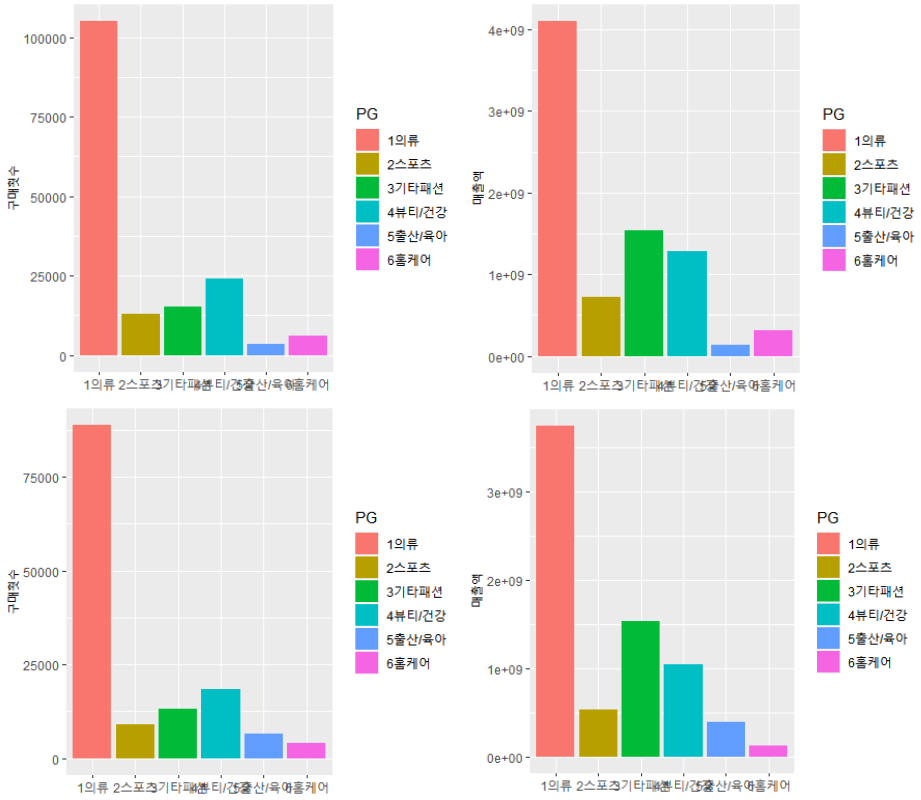




남성을 위한 서비스제안

```
pcm2_M40<-pcm2%>%filter(CLNT_GENDER=="M", CLNT_AGE==40)
ggplot(pcm2_M40%>%count(PG), aes(PG, n,fill=PG))+geom_bar(stat="identity")+xlab(NULL)+ylab("구매횟수")

pcm2_M40_rev<-pcm2_M40%>%group_by(PG)%>%summarize(REV_PG=sum(PD_BUY_TOTAL))
ggplot(pcm2_M40_rev,aes(PG,REV_PG,fill=PG))+geom_bar(stat="identity")+ylab("매출액")+xlab(NULL)
ggplot(pcm2_M30%>%count(PG), aes(PG, n,fill=PG))+geom_bar(stat="identity")+xlab(NULL)+ylab("구매횟수")
pcm2_M30_rev<-pcm2_M30%>%group_by(PG)%>%summarize(REV_PG=sum(PD_BUY_TOTAL))
ggplot(pcm2_M30_rev,aes(PG,REV_PG,fill=PG))+geom_bar(stat="identity")+ylab("매출액
```

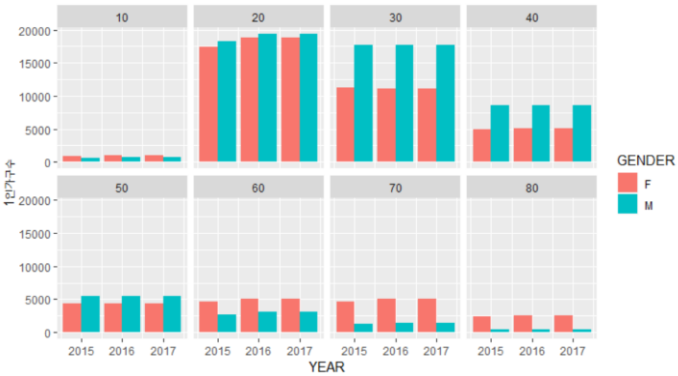
")+xlab(NULL)



"나혼자산다" 1인가구 맞춤컬렉션

연도별 행정구역별 1인가구 추이

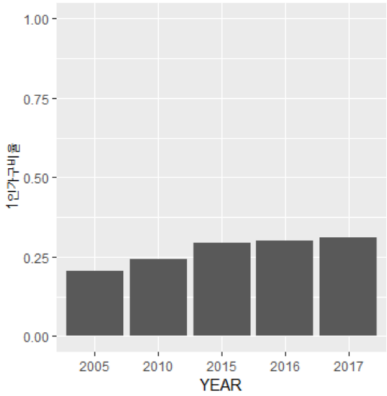
```
op1516<-union(op2015,op2016)
op151617<-union(op1516,op2017)
op151617<-op151617%>%arrange(YEAR,REGION,AGE)
op151617%>%ggplot(aes(YEAR,ALONE_CNT,fill=GENDER))+geom_col(position="dodge")+facet_wrap(~AGE,ncol=4)+scale_x_continuous(breaks=c(2015,2016,2017))+ylab("1인가구수")
```



연도별 1인가구 비율 추이

opt_chg%>%ggplot(aes(YEAR,ALONE_PCT))+geom_histogram(stat="identity")+scale_x_discrete(breaks=c(2005,2010,2015,2016,2017))+scale_y_continuous(limits=c(0,1))+ylab("1인가구비율")

```
> opt_chg
# A tibble: 5 x 4
  TOT_CNT ALONE_CNT  YEAR ALONE_PCT
    <int>     <int> <dbl>     <dbl>
1 3813260   1180540  2017     0.310
2 3784705   1138860  2016     0.301
3 3504297    854606  2010     0.244
4 3309890    675739  2005     0.204
5 3784490   1115744  2015     0.295
```



"일·가정 양립제도" 데이터 탐색

먼저, 자료정리

육아똑똑지수 파생변수 산출방법

가중치: 여자 각각의 "알고있다%"를 총합으로 나눔

가중치를 남자 각각의 "알고있다%"에 곱하여 더함.

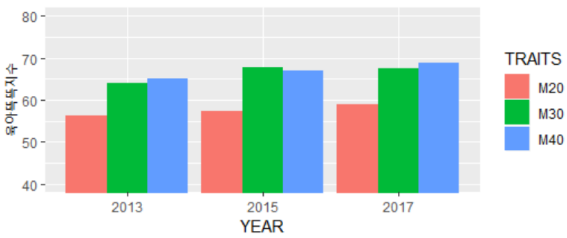
<parenting2013.csv정리본>

	A	B	C	D	E	F	G	H
1	TRAITS	출산(전후)휴가제 -	배우자출산휴가제	육아휴직제	육아기 근로시간 단축제	직장보육지원	가족돌봄휴직제	육아똑똑지수
2	M20	75.2	64.6	65.4	27	44.8	22.8	56.32068465
3	F20	81.1	66.4	75.9	40.4	51.2	29.7	
4	M30	82.2	75	74.3	36.6	54.2	26.6	63.89821518
5	F30	86.7	78.3	82.7	50.2	64.5	35.4	
6	M40	80.8	74	75.4	38.6	58	32.2	65.10352304
7	F40	82	71.9	77.4	42.5	59.5	35.7	

육아똑똑지수 연도별 추세확인

parenting2013<-read.csv("parenting2013.csv")

```
parenting2015<-read.csv("parenting2015.csv")
parenting2017<-read.csv("parenting2017.csv")
parenting2013<-as_tibble(parenting2013)
parenting2015<-as_tibble(parenting2015)
parenting2017<-as_tibble(parenting2017)
baby_smart13<-parenting2013%>%filter(TRAITS%in%c("M20","M30","M40"))%>%select(TRAITS,육아똑똑지수)
baby_smart15<-parenting2015%>%filter(TRAITS%in%c("M20","M30","M40"))%>%select(TRAITS,육아똑똑지수)
baby_smart17<-parenting2017%>%filter(TRAITS%in%c("M20","M30","M40"))%>%select(TRAITS,육아똑똑지수)
baby_smart13<-baby_smart13%>%mutate(YEAR=2013)
baby_smart15<-baby_smart15%>%mutate(YEAR=2015)
baby_smart17<-baby_smart17%>%mutate(YEAR=2017)
babysmart1315<-union(baby_smart13,baby_smart15)
babysmart131517<-union(babysmart1315,baby_smart17)
babysmart131517$TRAITS<-as.character(babysmart131517$TRAITS)
babysmart131517$YEAR<-as.character(babysmart131517$YEAR)
babysmart131517%>%ggplot(aes(YEAR,육아똑똑지수,fill=TRAITS))+geom_col(position="dodge")+scale_x_discrete(breaks=c(2013,2015,2017))+coord_cartesian(ylim=c(40,80))
```




```
> babysmart131517%>%arrange(TRAITS,YEAR)
# A tibble: 9 x 3
  TRAITS 육아똑똑지수 YEAR
  <chr>         <dbl> <chr>
1 M20            56.3 2013
2 M20            57.3 2015
3 M20            59.0 2017
4 M30            63.9 2013
5 M30            67.7 2015
6 M30            67.5 2017
7 M40            65.1 2013
8 M40            67.0 2015
9 M40            68.7 2017
```

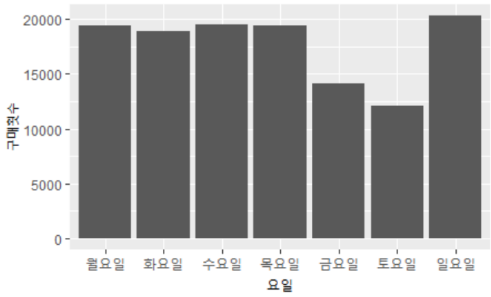
babysmart131517%>%arrange(TRAITS,YEAR)

아빠들의 육아제품 구매도를 높이자!

남자들의 요일별 육아용품 구매횟수

product1<-
product%>%group_by(CLNT_ID,SESS_ID)%>%summarize(PD_BUY_TOTCT=sum(PD_BUY_CT))

```
product1<-inner_join(product1,product)%>%select(CLNT_ID,SESS_ID,PD_BUY_TOTCT,PD_C)

pc<-inner_join(product1,custom)%>%select(CLNT_ID,SESS_ID,PD_C,PD_BUY_TOTCT,CLNT_GENDER)

pcm<-
left_join(pc,master)%>%select(CLNT_ID,SESS_ID,PD_C,PD_BUY_TOTCT,CLNT_GENDER,CLAC1_NM)

pcms<-
inner_join(pcm,session)%>%select(CLNT_ID,SESS_ID,PD_C,PD_BUY_TOTCT,CLNT_GENDER,CLAC1_NM,
SESS_DAY)

pcms1<-pcms%>%filter(CLNT_GENDER=="M",CLAC1_NM%in%c("유아동의류","출산/육아용품"))

pcms2<-
transform(pcms1,SESS_DAY=factor(SESS_DAY,levels=c("월요일","화요일","수요일","목요일","금요일","
토요일","일요일")))

pcms2%>%ggplot(aes(SESS_DAY,PD_BUY_TOTCT))+geom_col()+xlab("요일")+ylab("구매횟수")
```



금,토 가장 많이 팔리는 CLAC3_NM 육아제품

(앞에 만든 pcm 에 이어,,)

```
pcm<-
left_join(pc,master)%>%select(CLNT_ID,SESS_ID,PD_C,PD_BUY_TOTCT,CLNT_GENDER,CLAC1_NM,CLA
C3_NM)

pcms<-
inner_join(pcm,session)%>%select(CLNT_ID,SESS_ID,PD_C,PD_BUY_TOTCT,CLNT_GENDER,CLAC1_NM,
CLAC3_NM,SESS_DAY)

pcms1<-pcms%>%filter(CLNT_GENDER=="M",CLAC1_NM%in%c("유아동의류","출산/육아용품"))
```

fri_sat<-
fri_sat%>%group_by(CLAC3_NM)%>%summarize(sum=sum(PD_BUY_TOTCT))%>%arrange(desc(sum))

otherday<-pcms1%>%filter(SESS_DAY%in%c("월요일","화요일","수요일","목요일","일요일"))

otherday<-
otherday%>%group_by(CLAC3_NM)%>%summarize(sum=sum(PD_BUY_TOTCT))%>%arrange(desc(sum))

```
> fri_sat
# A tibble: 73 x 2
   CLAC3_NM              sum
   <fct>               <dbl>
 1 여아티셔츠/탑        3019
 2 영유아티셔츠/탑      2843
 3 남아바지             2579
 4 남아티셔츠/탑        2318
 5 유아용화장품         1225
 6 출산/신생아용품세트  1171
 7 여아패딩             1144
 8 영유아바지           1007
 9 남아의류세트          994
10 영유아원피스          747
# ... with 63 more rows
```

```
> otherday
# A tibble: 73 x 2
   CLAC3_NM              sum
   <fct>               <dbl>
 1 여아티셔츠/탑       11930
 2 영유아티셔츠/탑     10794
 3 남아바지             8016
 4 남아티셔츠/탑        7890
 5 출산/신생아용품세트  5411
 6 유아용화장품         4434
 7 영유아바지           4329
 8 남아의류세트         3610
 9 유아목욕용품         2667
10 유아용물티슈         2645
# ... with 63 more rows
```

서비스 아이디어: '월화수목일'에 적게 팔리지만 '금토'에 많이 팔리는 육아제품들을 뽑음. 따라서 뽑은 육아제품들을 '금토'에 구매하면 엘포인트 2 배 적립이라는 혜택을 줌.

이 서비스는 '금토' 남성들의 육아제품 구매횟수가 적으므로, 다른 요일과 다르게 많이 팔리고 있는 상품을 구매하도록 한다면, 신규고객유입을 얻을 수 있다. 따라서 상대적으로 구매횟수가 적은 '금토'에도 매출을 늘릴 수 있는 방안이다.

그러나 위의 분석을 진행해보니, 모든 요일에 팔리는 육아제품 순위가 비슷한 것을 확인함.

따라서 서비스는 제안하지 않기로 결정.(PPT 수록 제외)